

Computer Networks

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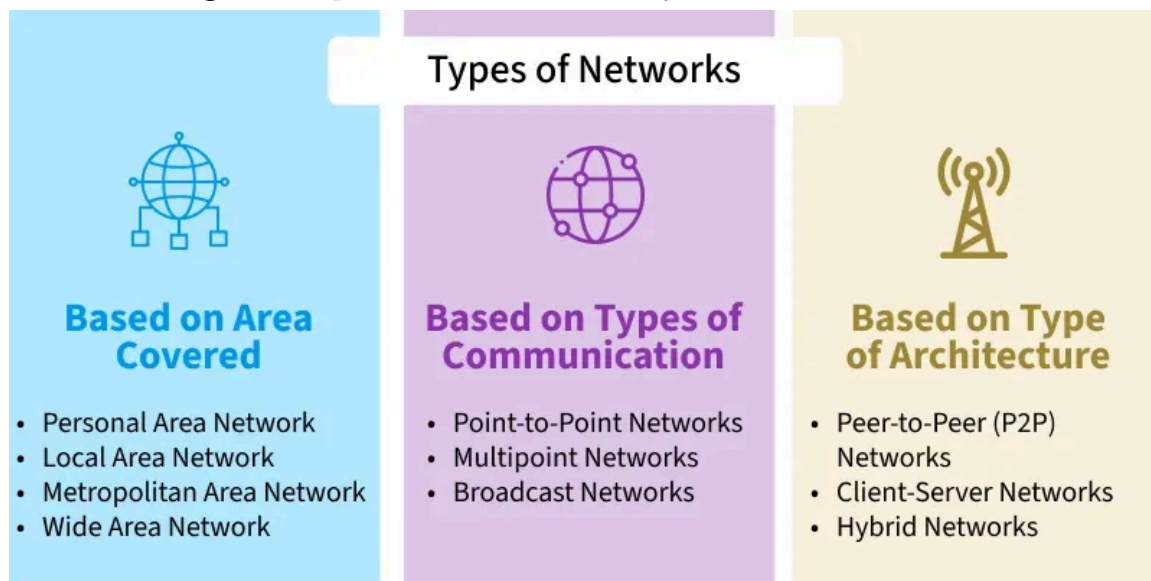
- TCP 3-way handshake
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1.1 Network – Definition

A **computer network** is a collection of interconnected devices (such as computers, servers, or phones) that communicate with each other to **share data, resources, and services** using communication protocols.

1.2 Types of Network - LAN, WAN and MAN

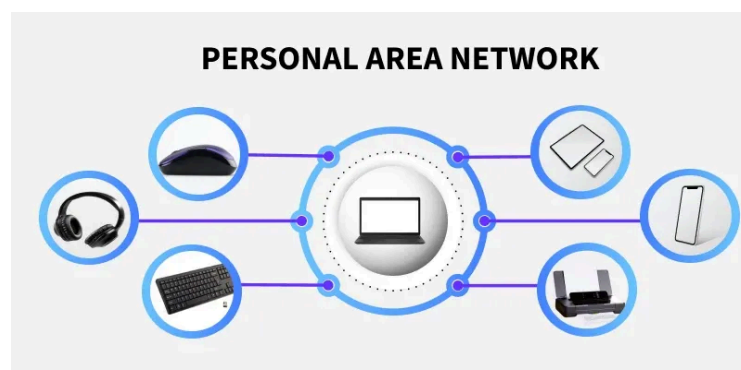
The Network allows computers to connect and communicate with different computers via any medium. LAN, MAN, and WAN are the three major types of networks designed to operate over the area they cover.



The different types of networks can be categorised based on the area they cover (ranging from personal connections to global networks), the type of communication they use, and their architectural design.

1. Personal Area Network (PAN)

A PAN is a short-range network that connects personal devices like smartphones, tablets, and computers. It typically covers less than 10 meters (about 33 feet) and usually uses wireless technologies such as Bluetooth. PAN is smaller than other networks like LAN or WAN and is mainly used for data sharing between a few devices.



Advantages:

- Allows for easy communication between personal devices in close proximity.
- Uses wireless technology, which eliminates the need for wires and cables.
- PANs are designed to be energy efficient, which means that devices can communicate with each other without draining their batteries quickly.

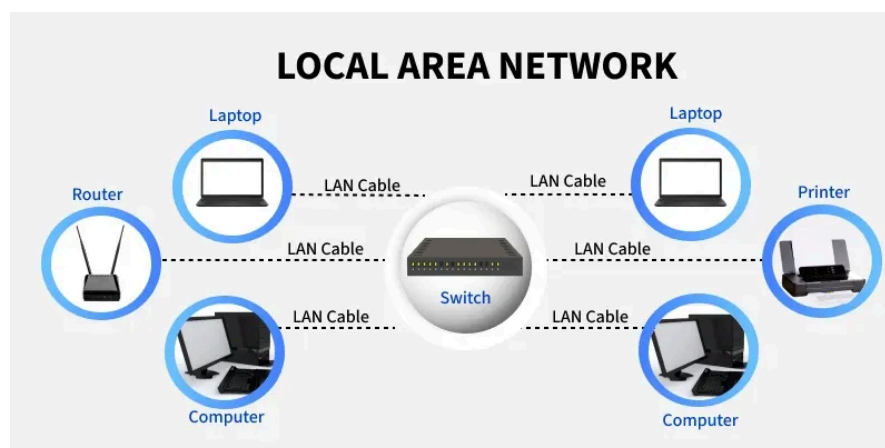
Disadvantages:

- Limited coverage area.
- PANs have limited bandwidth, so they are not suitable for large-scale or high-speed data transfer.
- May experience interference from other wireless devices.

2. Local Area Network (LAN)

A LAN connects computers and devices within a small area like a home, office, school, or hospital. It usually uses switches, routers, and private IP addresses. LANs are high-speed, inexpensive to set up, and easy to maintain.

- Coverage: Up to 2 km (limited area).
- Speed: Earlier 4–16 Mbps; now 100–1000 Mbps.
- Medium: Mostly wired (Ethernet cables, twisted-pair, coaxial), though wireless can also be used.
- Example: Students playing a multiplayer game in the same room without internet.



Advantages:

- Provides fast data transfer rates and high-speed communication.
- Easy to set up and manage.
- Can be used to share peripheral devices such as printers and scanners.

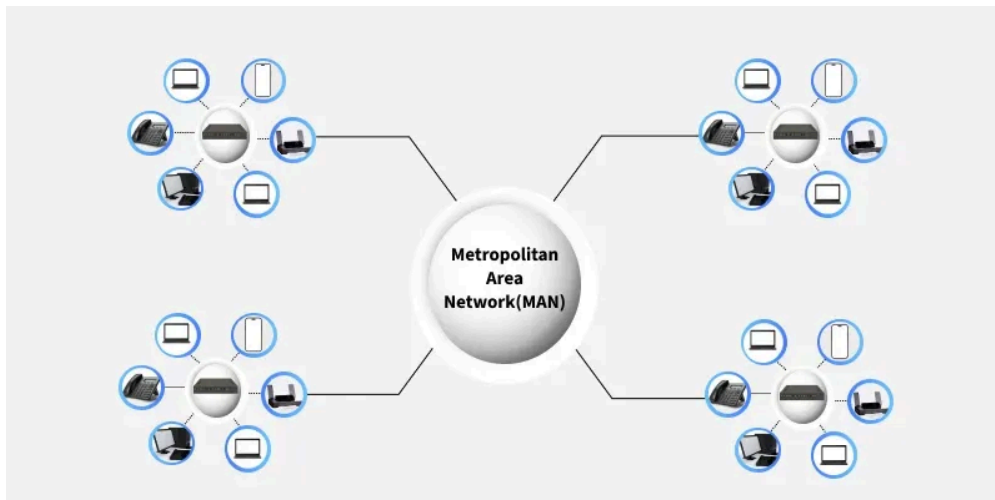
Disadvantages:

- Limited geographical coverage.

- Limited scalability and may require significant infrastructure upgrades to accommodate growth.
- May experience congestion and network performance issues with increased usage.

3. Metropolitan Area Network (MAN)

A MAN spans 5–50 km, covering more area than a LAN but less than a WAN. It connects computers across a city or between nearby cities. MANs provide high-speed connectivity (in Mbps), can act as ISPs, and are useful for organizations needing fast communication. However, they are costly, complex to design, and harder to maintain.



Advantages:

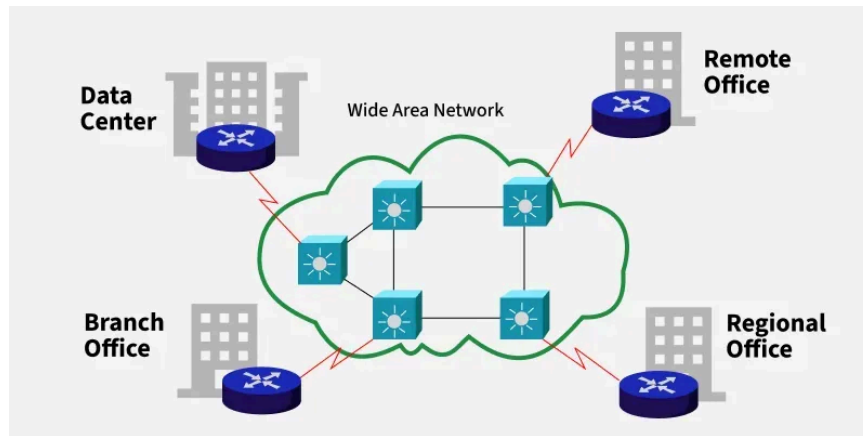
- Provides high-speed connectivity over a larger geographical area than LAN.
- Can be used as an ISP for multiple customers.
- Offers higher data transfer rates than WAN in some cases.

Disadvantages:

- Can be expensive to set up and maintain.
- May experience congestion and network performance issues with increased usage.
- May have limited fault tolerance and security compared to LANs.

4. Wide Area Network (WAN)

A WAN covers large geographical areas (above 50 km), often connecting multiple LANs through telephone lines, radio waves, or satellites. It can be private (for organizations) or public (like the internet). WANs offer high-speed communication but are costly to set up and maintain.



Advantages:

- Covers large geographical areas and can connect remote locations.
- Provides connectivity to the internet.
- Offers remote access to resources and applications.

Disadvantages:

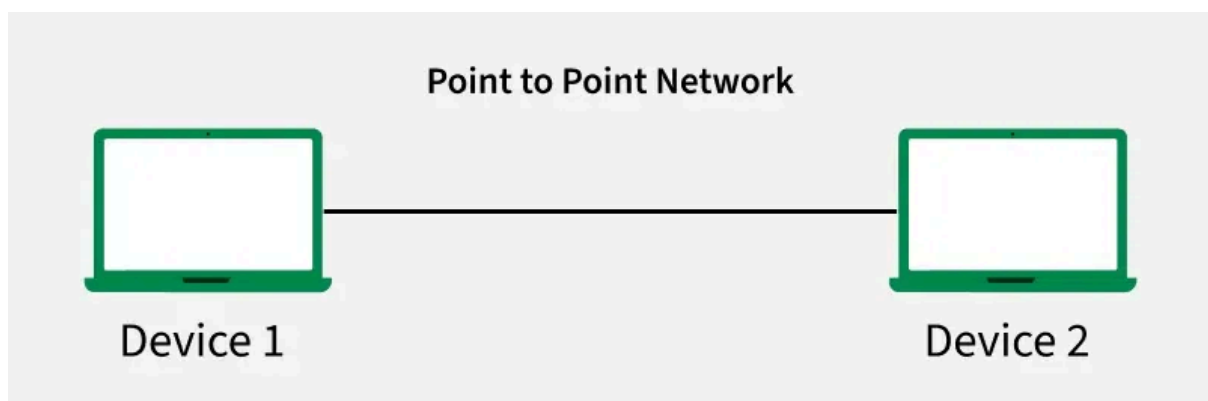
- Can be expensive to set up and maintain.
- Offers slower data transfer rates than LAN or MAN.
- May have lower fault tolerance and security compared to LANs.

Based on Types of Communication

Networks can also be divided based on the types of communication they use:

1. Point-to-Point

A Point-to-Point network creates a direct connection between two devices (e.g., computer and printer). Data is transmitted without intermediaries, using either wired (cables) or wireless (infrared, radio) links. It is simple, dedicated, and commonly used for two-device communication.



Advantages:

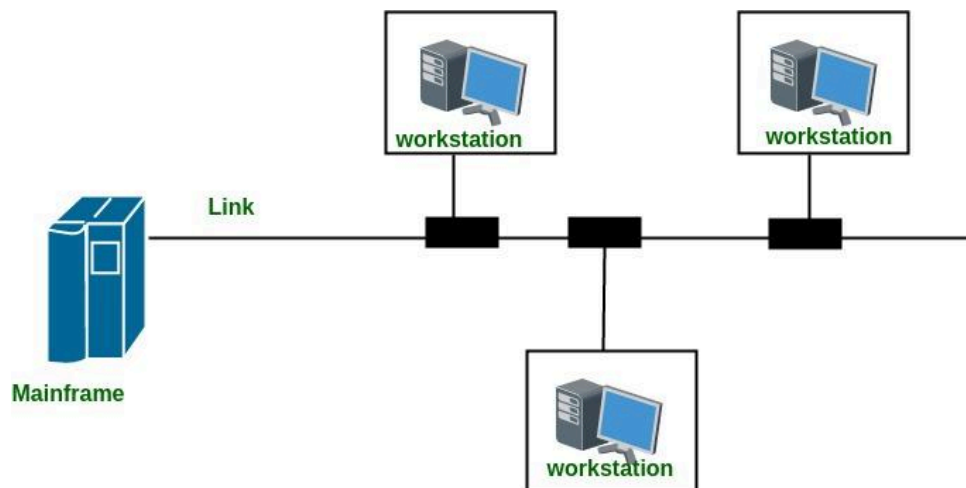
- Provides a dedicated and direct connection, ensuring reliable communication between two devices.
- Simple to set up and configure due to the limited number of devices involved.
- Offers low latency as there are no intermediate devices or complex routing.

Disadvantages:

- Not scalable, as adding more devices requires additional dedicated links.
- May require physical cables for wired connections, which can limit flexibility.
- Lacks fault tolerance; if the direct link fails, communication is disrupted.

2. Multipoint Networks

A multipoint network allows more than two devices to share a single communication link. The channel capacity is either shared at the same time (spatially) or in turns (temporally).



Advantages:

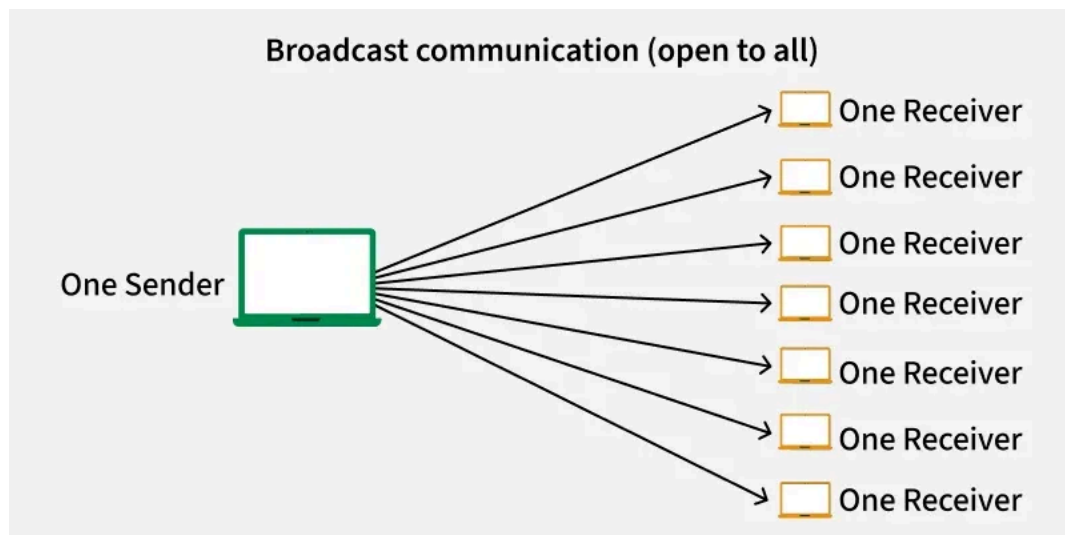
- Allows multiple devices to share a single communication channel, reducing the need for multiple dedicated links.
- More cost-effective than point-to-point networks for connecting multiple devices.
- Supports dynamic communication, where devices can join or leave the network easily.

Disadvantages:

- Shared bandwidth can lead to congestion and reduced performance when many devices are active.
- Potential for interference or collisions in spatially shared connections.
- Security risks may increase as multiple devices access the same communication medium.

3. Broadcast Networks

Broadcast networks use a single communication channel where one sender transmits data that can be received by multiple parties simultaneously. In this setup, data travels in one direction, from the sender to all potential receivers. A common example is a radio station, where the station broadcasts signals that can be picked up by any radio receiver within range. Another example is a television network broadcasting to multiple viewers



Advantages:

- Efficient for distributing data to a large number of recipients simultaneously.
- Simple to implement for one-to-many communication scenarios.
- Reduces the need for individual connections to each receiver, saving resources.

Disadvantages:

- Data is sent in one direction, limiting two-way communication.
- No guarantee that all intended recipients receive the data reliably.
- Can lead to bandwidth wastage if not all receivers need the transmitted data.

1.3 Network Topology

Network Topology refers to the arrangement and organization of devices in a computer network. It defines how components are structured and how communication is established between them.

- ❖ Describes the overall structure of a network
- ❖ Helps in network design and planning
- ❖ Influences performance and reliability

- ❖ Includes Physical Topology, which shows the actual layout of devices and cables
- ❖ Includes Logical Topology, which defines how data moves between devices

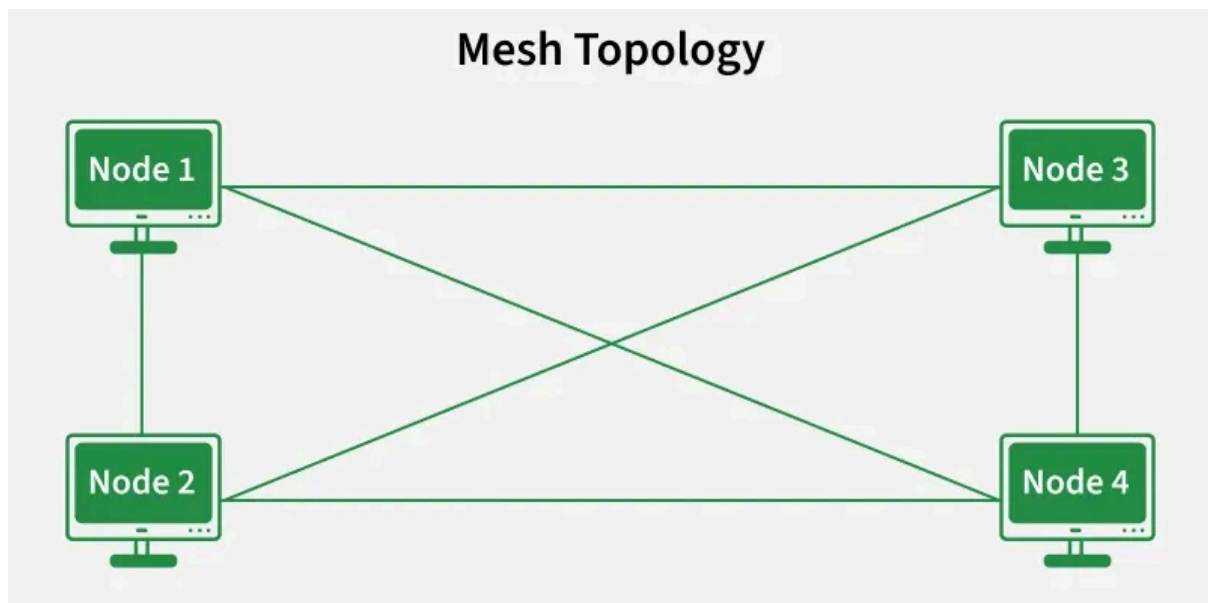
Types of Network Topology

The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as Network Topology. Below are various network topologies that include:

1. Mesh Topology

Mesh Topology is a network structure in which each device is connected to multiple other devices, creating several communication paths within the network. It provides high redundancy and reliability.

- Devices are interconnected with dedicated links
- Supports multiple paths for data transmission
- Can be full mesh or partial mesh



There are two types of Mesh topologies:

1. Full Mesh Topology:

In a full mesh network, every node is directly connected to all other nodes. If there are N nodes, each node has $(N - 1)$ connections. This provides maximum redundancy and reliability.

2. Partial Mesh Topology:

In a partial mesh network, only selected nodes are interconnected. Not every node is directly connected to all others. This approach reduces cost and complexity while still maintaining reasonable reliability.

Advantages

- High fault tolerance
- Reliable data delivery
- Enhanced security due to dedicated links
- Suitable for critical network systems

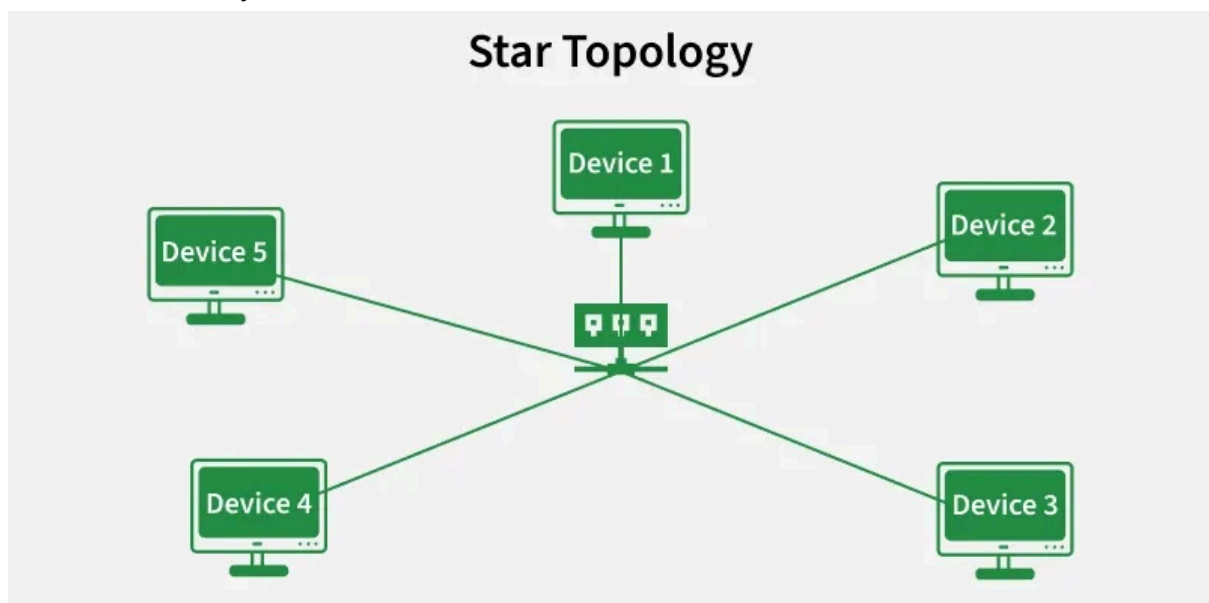
Disadvantages

- Expensive to implement
- Requires large amount of cabling
- Complex installation process
- Difficult to manage in large networks

2. Star Topology

Star Topology is a network layout in which all devices are connected to a central device such as a switch or hub. All communication between devices passes through this central point.

- Each device has a dedicated connection to the central node
- Central device controls data transmission
- Commonly used in Local Area Networks (LANs)



Advantages

- Easy to install and manage
- Simple fault detection and troubleshooting
- Failure of one device does not affect others
- Easy to add or remove devices

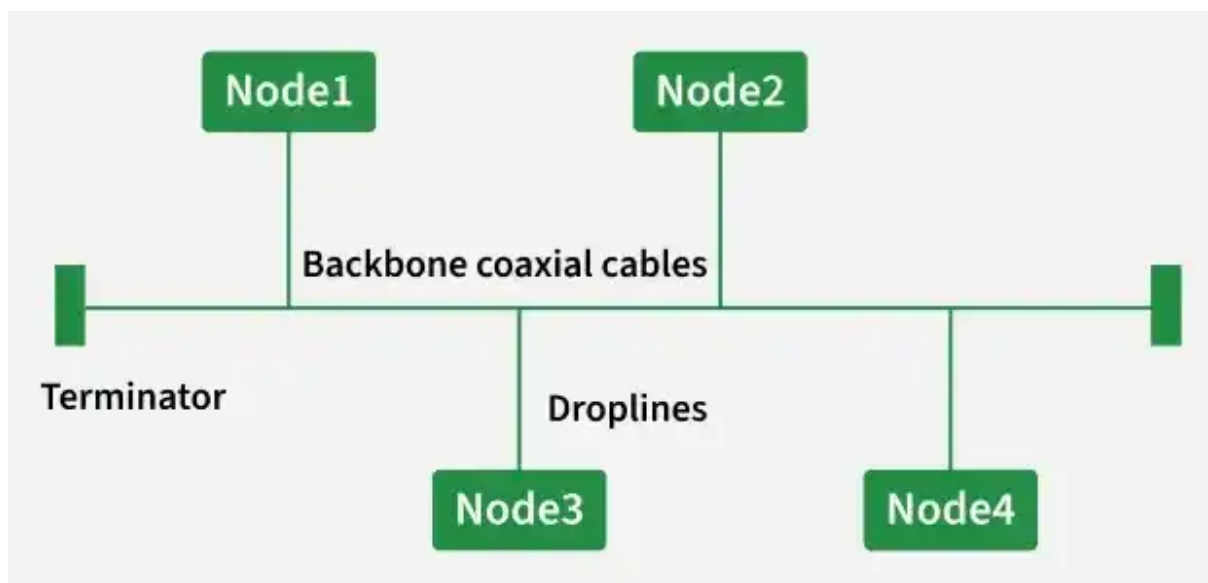
Disadvantages

- Failure of central device stops the entire network
- Requires more cabling than bus topology
- Higher installation cost
- Performance depends on central device capacity

3. Bus Topology

Bus Topology is a network structure in which all devices are connected to a single main communication cable called the backbone. Data sent by any device travels along this common cable and is received by the intended device.

- Uses a single backbone cable
- All devices share the same communication channel
- Terminators are placed at both ends of the cable



Advantages

- Simple and easy to implement
- Requires less cable compared to star topology
- Cost-effective for small networks
- Suitable for temporary network setups

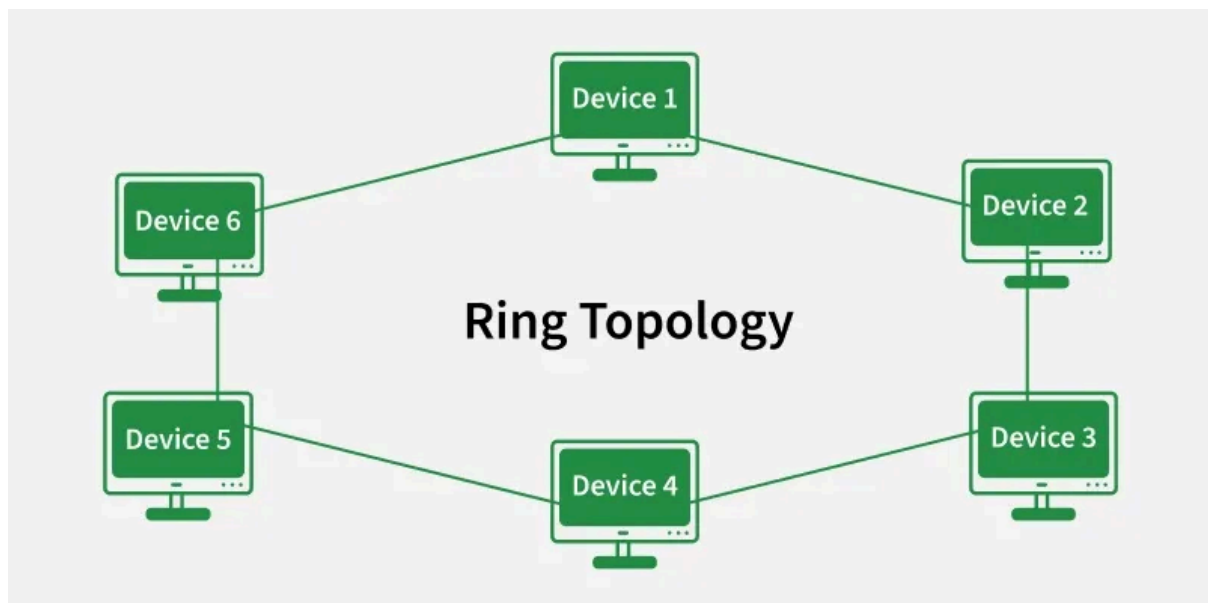
Disadvantages

- Failure of backbone cable stops the entire network
- Difficult to troubleshoot issues
- Performance decreases with high traffic
- Limited scalability for large networks

4. Ring Topology

Ring Topology is a network configuration in which each device is connected to two other devices, forming a closed circular path. Data travels around the ring in a fixed direction until it reaches its destination.

- Forms a continuous circular structure
- Each device acts as a repeater
- Data passes through intermediate devices



Advantages

- Reduces chances of data collision
- Provides equal access to all devices
- Performs well under consistent traffic load
- Suitable for orderly data transmission

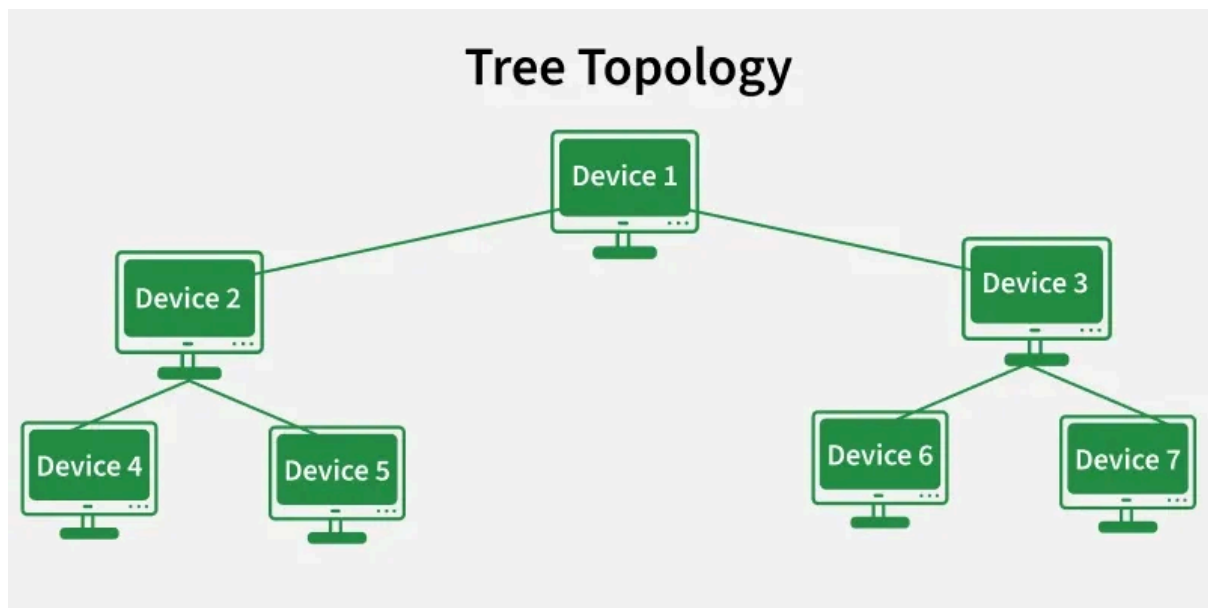
Disadvantages

- Failure of one device can disrupt communication
- Troubleshooting can be difficult
- Adding or removing devices affects the network
- Slower compared to some modern topologies

5. Tree Topology

Tree Topology is a hierarchical network structure in which multiple star networks are connected to a central backbone cable. It combines features of both bus and star topologies and organizes devices in a parent–child arrangement.

- Follows a hierarchical structure
- Consists of root node and branch nodes
- Uses a backbone cable for connection
- Suitable for large networks



Advantages

- Easy to expand by adding new branches
- Supports structured network management
- Fault isolation is simpler
- Scalable for growing organizations

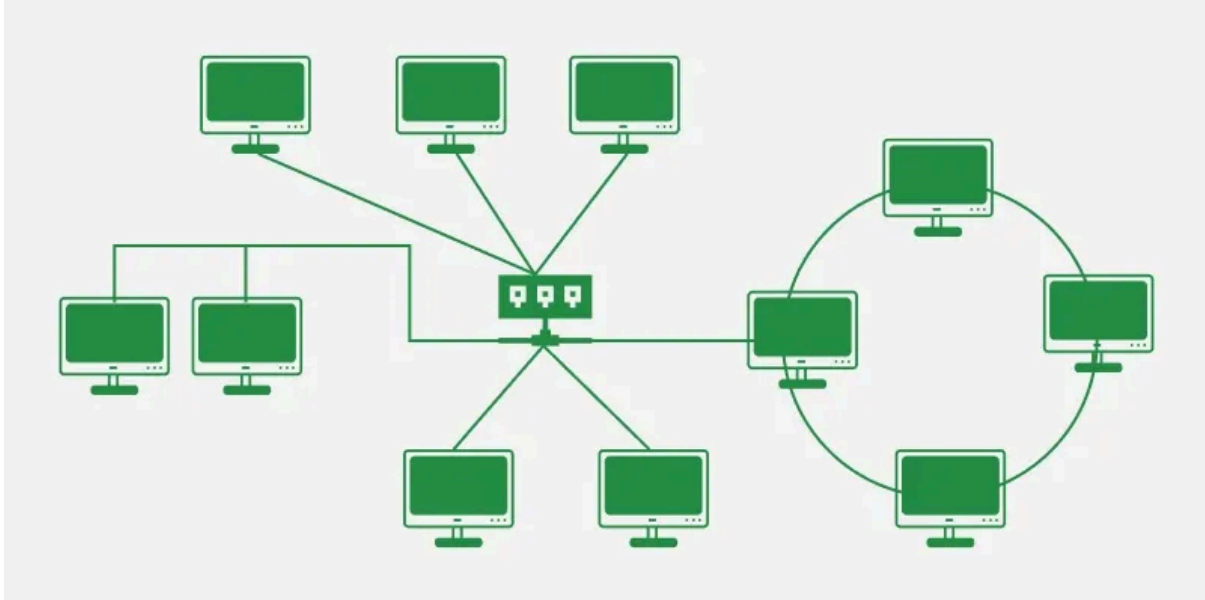
Disadvantages

- Backbone failure affects entire network
- Requires more cabling
- Configuration is complex
- Maintenance can be challenging

6. Hybrid Topology

Hybrid Topology is a network structure that combines two or more different types of topologies into a single system. It is designed to meet specific organizational requirements by integrating the strengths of multiple network layouts.

- Formed by combining different topologies
- Customized according to network needs
- Common in large enterprise environments
- Supports complex network structures



Advantages

- Flexible network design
- Scalable for future expansion
- Can improve reliability
- Allows optimization based on requirements

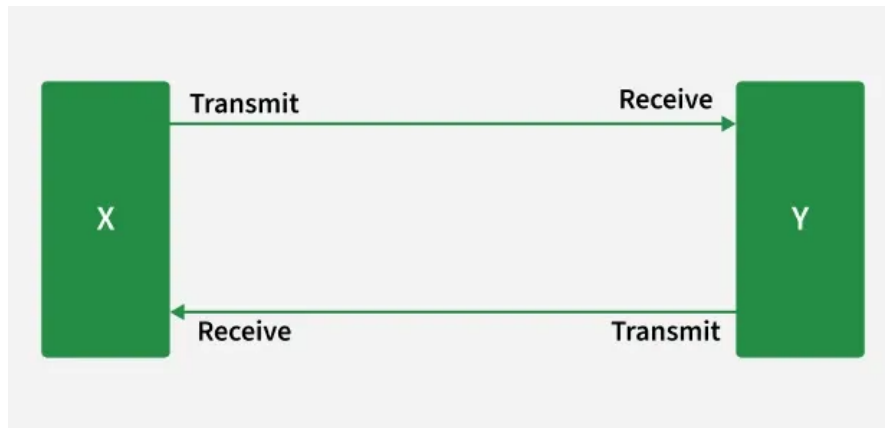
Disadvantages

- Expensive to implement
- Complex to design and manage
- Requires advanced hardware
- Maintenance can be difficult

7. Point to Point Topology

Point-to-Point Topology is a network configuration in which two devices are directly connected through a dedicated communication link. The connection is exclusive, meaning the entire bandwidth is used only between these two devices.

- Involves only two connected devices
- Uses a single dedicated link
- Data travels directly without intermediaries
- Simple network structure



Advantages

- High data transfer speed
- Secure communication channel
- Easy to configure
- Low chances of data collision

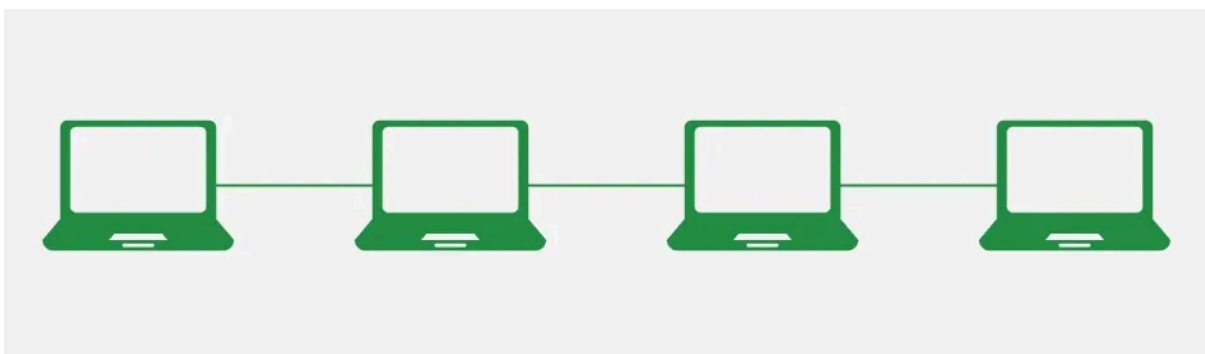
Disadvantages

- Not suitable for large networks
- Limited scalability
- Link failure stops communication
- Inefficient for multiple device connections

8. Daisy Chain Topology

Daisy Chain Topology is a network arrangement in which devices are connected sequentially, one after another, forming a linear chain. Each device is linked to the next device in the sequence.

- Devices are connected in series
- Data passes through intermediate devices
- Simple linear structure
- Often used in small or temporary networks



Advantages

- Easy to set up
- Requires less cabling than complex topologies
- Cost-effective for limited devices
- Simple expansion by adding devices at the end

Disadvantages

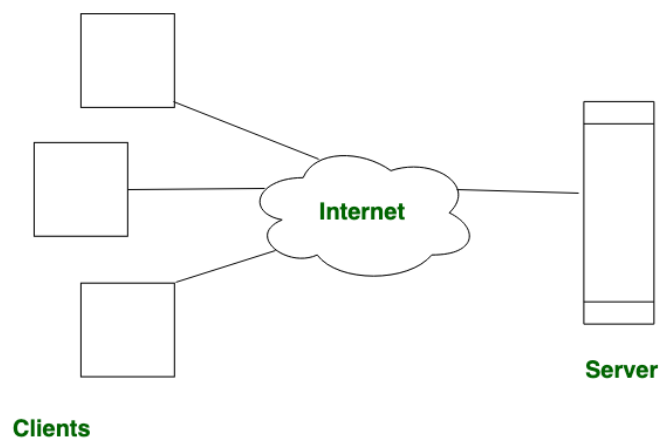
- Failure of one device can disrupt the chain
- Performance decreases as more devices are added
- Difficult to troubleshoot faults
- Not suitable for large-scale networks

1.4 Difference between Client-Server and Peer-to-Peer Network

In the world of network architecture, two fundamental models are widely utilized to structure data exchange and resource sharing. For the purpose of this discussion, two types of networks are available; the Client-Server Network and the Peer-to-Peer Network. All the models have their strengths, weaknesses, and appropriate applications that make them suitable for use. An understanding of these differences will assist in choosing suitable approaches for different networking requirements.

What is a Client-Server Network?

This model are broadly used network model. In the Client-Server Network, Clients and servers are differentiated, and Specific servers and clients are present. In Client-Server Network, a Centralized server is used to store the data because its management is centralized. In Client-Server Network, the Server responds to the services which is requested by the Client.



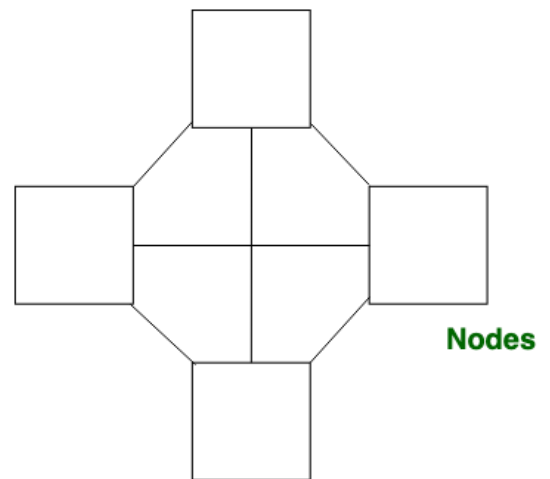
Client-Server Network Model

What is Peer-to-Peer Network?

This model does not differentiate the clients and the servers, In this each and every node is itself client and server. In Peer-to-Peer Network, Each and every node can do both request and respond for the services.

- ☐ Peer-to-peer networks are often created by collections of 12 or fewer machines. All of these computers use unique security to keep their data, but they also share data with every other node.

- ☐ In peer-to-peer networks, the nodes both consume and produce resources. Therefore, as the number of nodes grows, so does the peer-to-peer network's capability for resource sharing. This is distinct from client-server networks where an increase in nodes causes the server to become overloaded.
- ☐ It is challenging to give nodes in peer-to-peer networks proper security because they function as both clients and servers. A denial of service attack may result from this.
- ☐ The majority of contemporary operating systems, including Windows and Mac OS, come with software to implement peer



Peer-to-Peer Network Model

Difference Between Client-Server and Peer-to-Peer Network

Client-Server Network	Peer-to-Peer Network
In Client-Server Network, Clients and server are differentiated, Specific server and clients are present.	In Peer-to-Peer Network, Clients and server are not differentiated.
Client-Server Network focuses on information sharing.	While Peer-to-Peer Network focuses on connectivity.
In Client-Server Network, Centralized server is used to store the data.	While in Peer-to-Peer Network, Each peer has its own data.
In Client-Server Network, Server respond the services which is request by Client.	While in Peer-to-Peer Network, Each and every node can do both request and respond for the services.
Client-Server Network are costlier than Peer-to-Peer Network.	While Peer-to-Peer Network are less costlier than Client-Server Network.
Client-Server Network are more stable than Peer-to-Peer Network.	While Peer-to-Peer Network are less stable if number of peer is increase.
Client-Server Network is used for both small and large networks.	While Peer-to-Peer Network is generally suited for small networks with fewer than 10 computers.

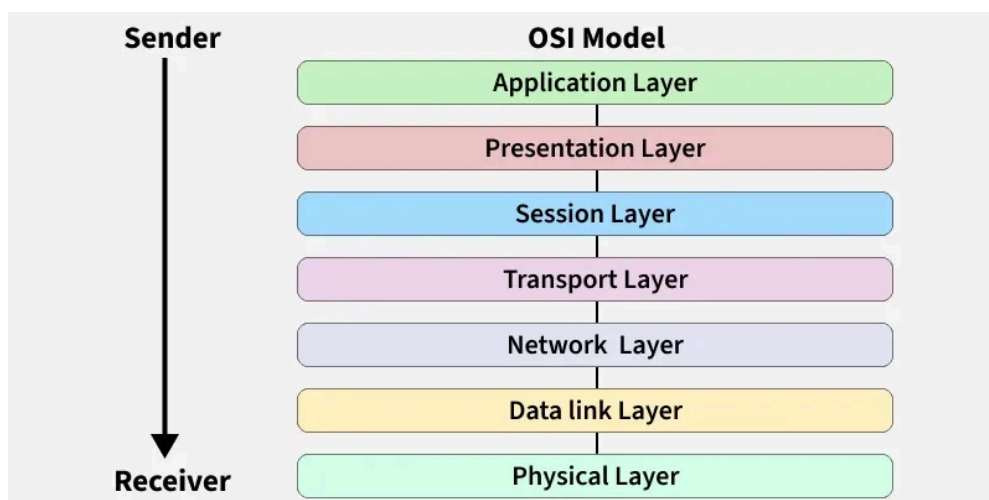
Conclusion

While Client-Server network has inherent merits which make it suitable for specific conditions, so too has the Peer-to-Peer network. The Client-Server model can be useful where there is the need to have centralized management and is more scalable than the Peer-to-Peer model which is suitable for scenarios where there is decentralization and direct sharing of the resources. Such a distinction might help in choosing the right type of a network model depending on the particular need and demand.

2. Network Models

2.1 What is OSI Model? - Layers of OSI Model

The OSI (Open Systems Interconnection) Model is a set of rules that explains how different computer systems communicate over a network. OSI Model was developed by the International Organization for Standardization (ISO). The OSI Model consists of 7 layers and each layer has specific functions and responsibilities. This layered approach makes it easier for different devices and technologies to work together.

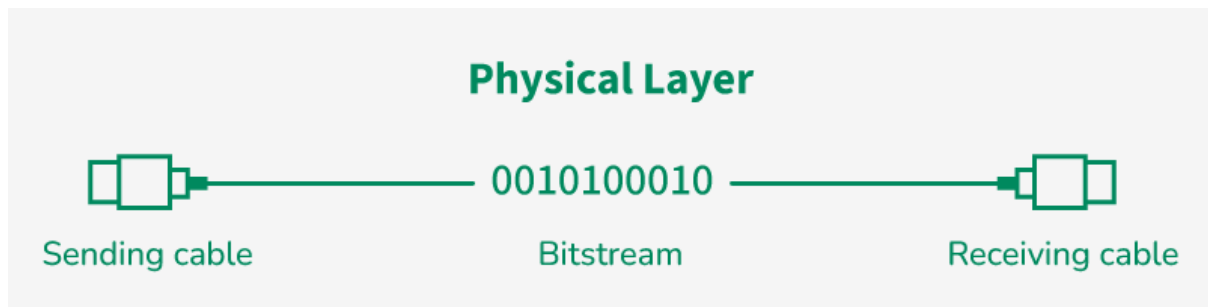


Note: OSI Model provides a clear structure for data transmission and managing network issues. The OSI Model is widely used as a reference to understand how network systems function.

Layer 1: Physical Layer

The lowest layer of the OSI reference model is the Physical Layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of bits.

- Physical Layer is responsible for transmitting individual bits from one node to the next.
- When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.
- Common physical layer devices are Hub, Repeater, Modem, and Cables.



Functions of the Physical Layer

- Bit Synchronization: The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- Bit Rate Control: The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- Physical Topologies: Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus topology, star topology, or mesh topology.
- Transmission Mode: Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full duplex.

Layer 2: Data Link Layer (DLL)

The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.

- When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.
- Packet in the Data Link layer is referred to as Frame. Switches and Bridges are common Data Link Layer devices.
- The packet received from the Network layer is further divided into frames depending on the frame size of the NIC (Network Interface Card). DLL also encapsulates Sender and Receiver's MAC address in the header.
- The Receiver's MAC address is obtained by placing an ARP (Address Resolution Protocol) request onto the wire asking, "Who has that IP address?" and the destination host will reply with its MAC address.

Sublayers of Data Link Layer

- Logical Link Control (LLC)
- Media Access Control (MAC)

Functions of the Data Link Layer

- Framing: Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- Physical Addressing: After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- Error Control: The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- Flow Control: The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- Access Control: When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

Layer 3: Network Layer

The Network Layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available.

- The sender and receiver's IP address are placed in the header by the network layer. Segment in the Network layer is referred to as Packet.
- Network layer is implemented by networking devices such as routers and switches.

Functions of the Network Layer

- Routing: The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- Logical Addressing: To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender and receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.

Layer 4: Transport Layer

The Transport Layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as Segments. It is responsible for the end-to-end delivery of the complete message.

- The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.
- Protocols used in Transport Layer are TCP, UDP NetBIOS, PPTP.
- At the sender's side, the transport layer receives the formatted data from the upper layers, performs Segmentation, and also implements Flow and error control to ensure proper data transmission.
- It also adds Source and Destination port number in its header and forwards the segmented data to the Network Layer.
- Generally, this destination port number is configured, either by default or manually.
- Example: when a web application requests a web server, it typically uses port number 80, because this is the default port assigned to web applications. Many applications have default ports assigned.
- At the Receiver's side, Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.

Functions of the Transport Layer

- Segmentation and Reassembly: This layer accepts the message from the (session) layer and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- Service Point Addressing: To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus, by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer

- Connection-Oriented Service
- Connectionless Service

Layer 5: Session Layer

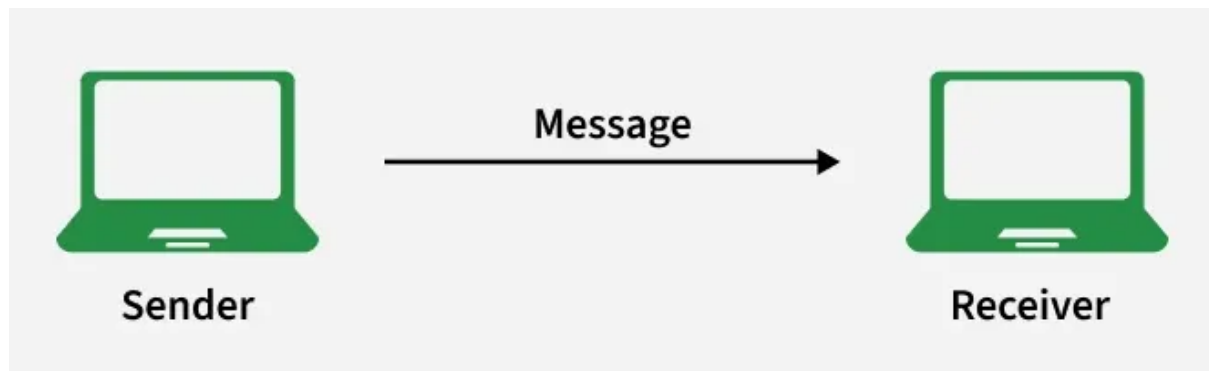
Session Layer in the OSI Model is responsible for the establishment of connections, management of connections, terminations of sessions between two devices. It also provides authentication and security. Protocols used in the Session Layer are NetBIOS, PPTP.

Functions of the Session Layer

- Session Establishment, Maintenance, and Termination: The layer allows the two processes to establish, use, and terminate a connection.
- Synchronization: This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely, and data loss is avoided.
- Dialog Controller: The session layer allows two systems to start communication with each other in half-duplex or full duplex.

Example

Let us consider a scenario where a user wants to send a message through some Messenger application running in their browser. The “Messenger” here acts as the application layer which provides the user with an interface to create the data. This message or so-called Data is compressed, optionally encrypted (if the data is sensitive), and converted into bits (0's and 1's) so that it can be transmitted.



Layer 6: Presentation Layer

The Presentation Layer is also called the Translation layer. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network. Protocols used in the Presentation Layer are TLS/SSL (Transport Layer Security / Secure Sockets Layer). JPEG, MPEG, GIF, are standards or formats used for encoding data, which is part of the presentation layer's role.

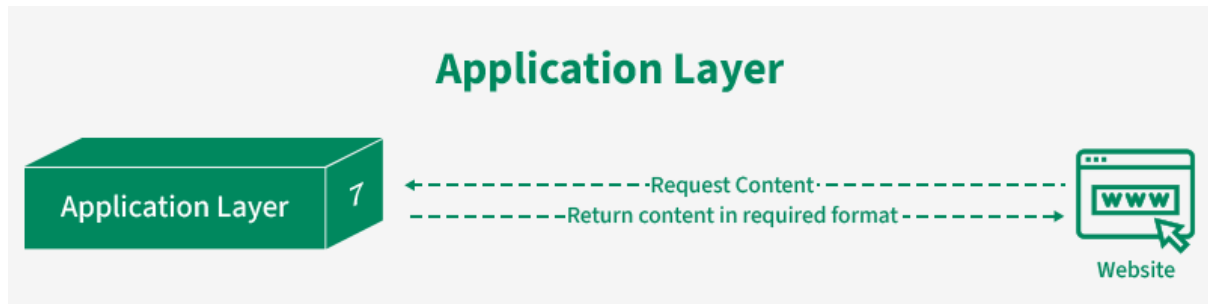
Functions of the Presentation Layer

- Translation: For example, ASCII to EBCDIC.
- Encryption/ Decryption: Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext, and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- Compression: Reduces the number of bits that need to be transmitted on the network.

Layer 7: Application Layer

At the very top of the OSI Reference Model stack of layers, we find the Application Layer which is implemented by the network applications. These applications produce the data to be transferred over the network.

- This layer also serves as a window for the application services to access the network and for displaying the received information to the user.
- Protocols used in the Application layer are SMTP, FTP, DNS, etc.



Functions of the Application Layer

The main functions of the application layer are given below.

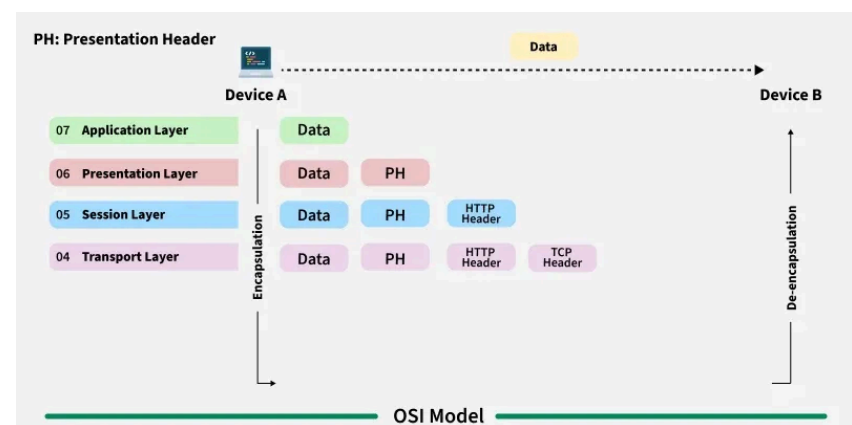
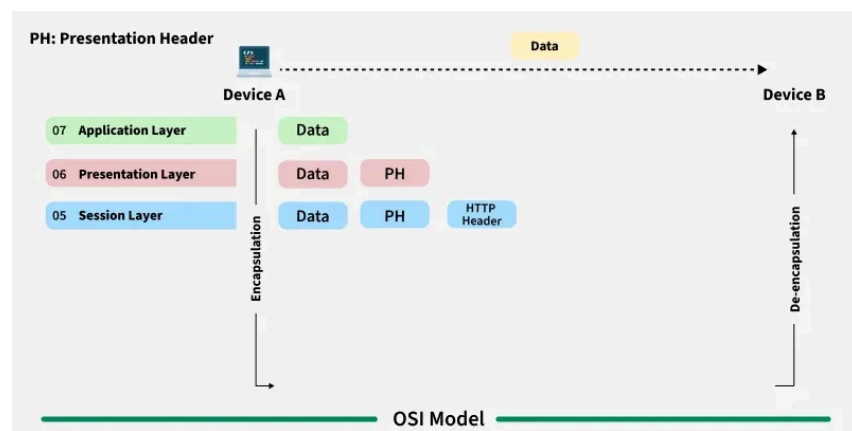
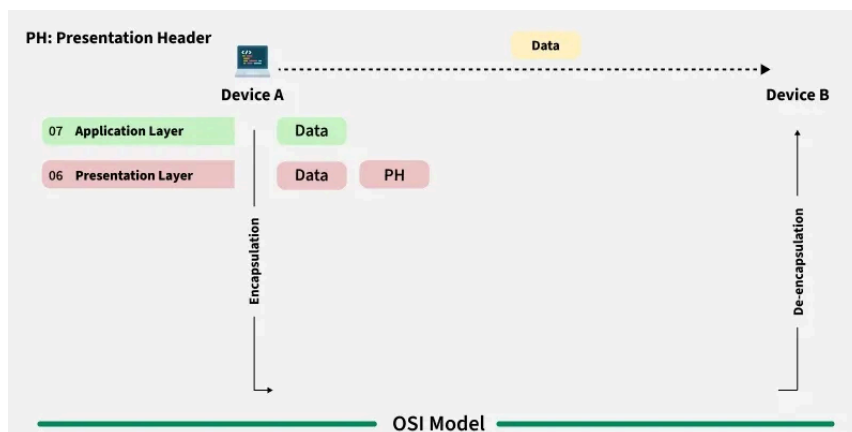
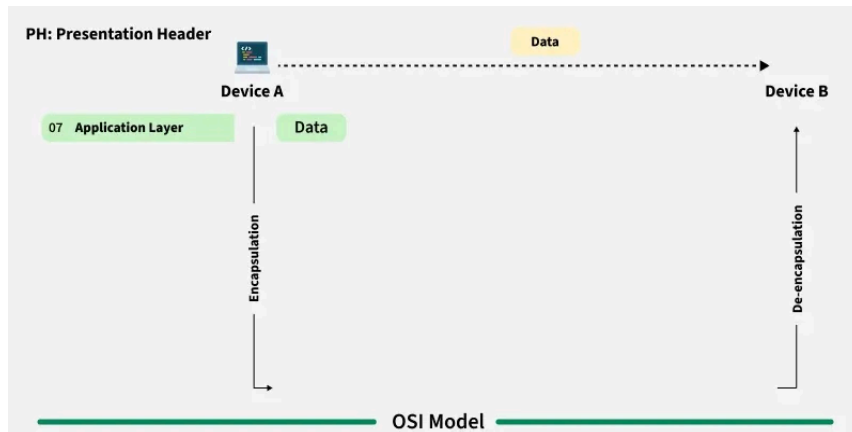
- Network Virtual Terminal (NVT): It allows a user to log on to a remote host.
- File Transfer Access and Management (FTAM): This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- Mail Services: Provide email service.
- Directory Services: This application provides distributed database sources and access for global information about various objects and services.

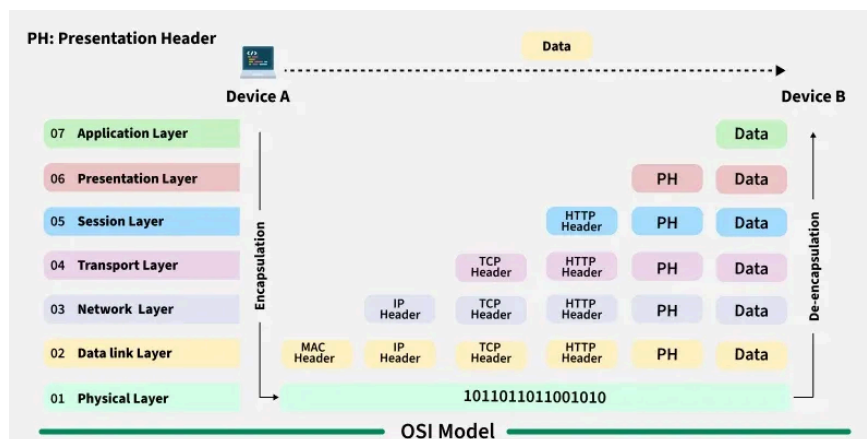
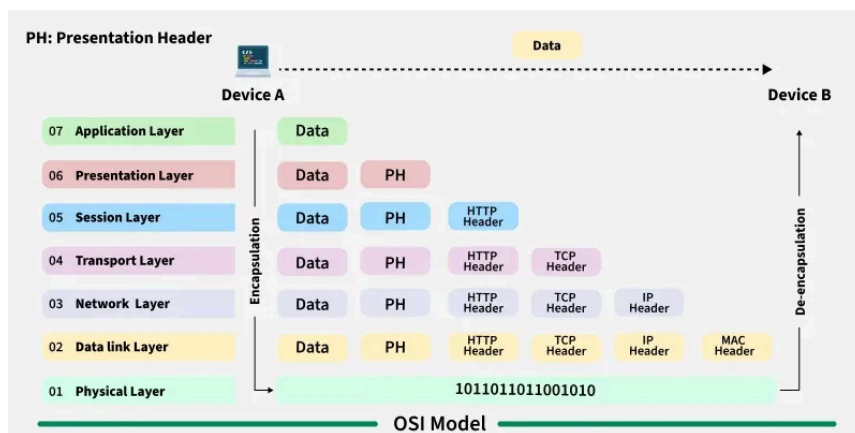
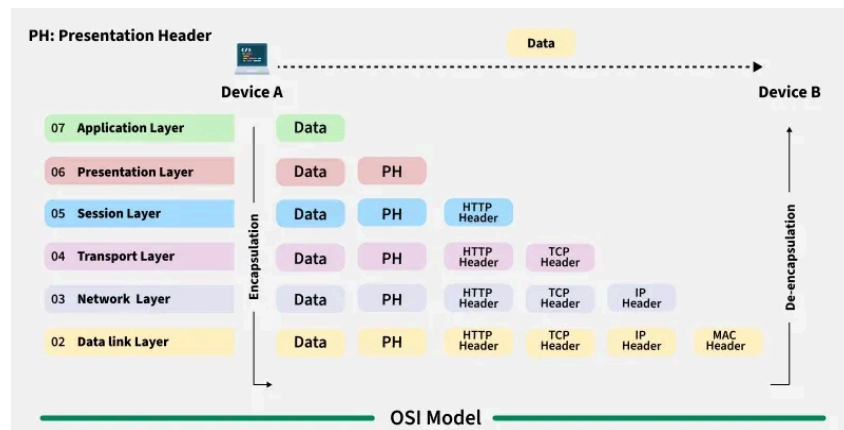
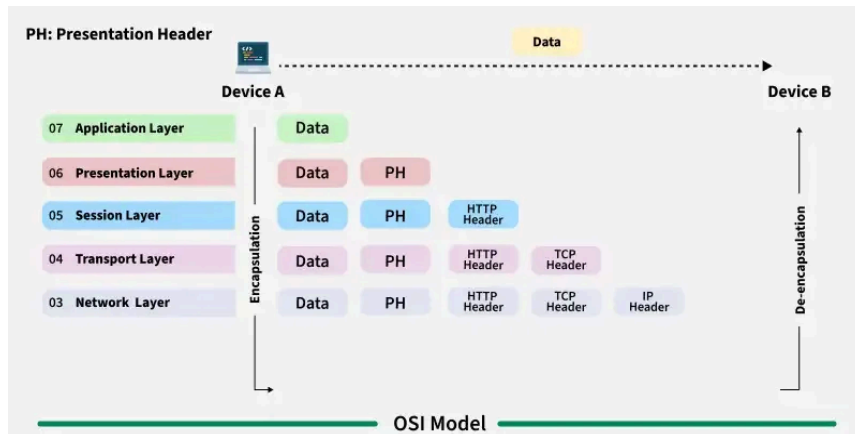
How Data Flows in the OSI Model?

When we transfer information from one device to another, it travels through 7 layers of OSI model. First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end. Data flows through the OSI model in a step-by-step process:

- ☐ Application Layer: Applications create the data.
- ☐ Presentation Layer: Data is formatted and encrypted.
- ☐ Session Layer: Connections are established and managed.
- ☐ Transport Layer: Data is broken into segments for reliable delivery.
- ☐ Network Layer: Segments are packaged into packets and routed.
- ☐ Data Link Layer: Packets are framed and sent to the next device.
- ☐ Physical Layer: Frames are converted into bits and transmitted physically.

Each layer adds specific information to ensure the data reaches its destination correctly, and these steps are reversed upon arrival.

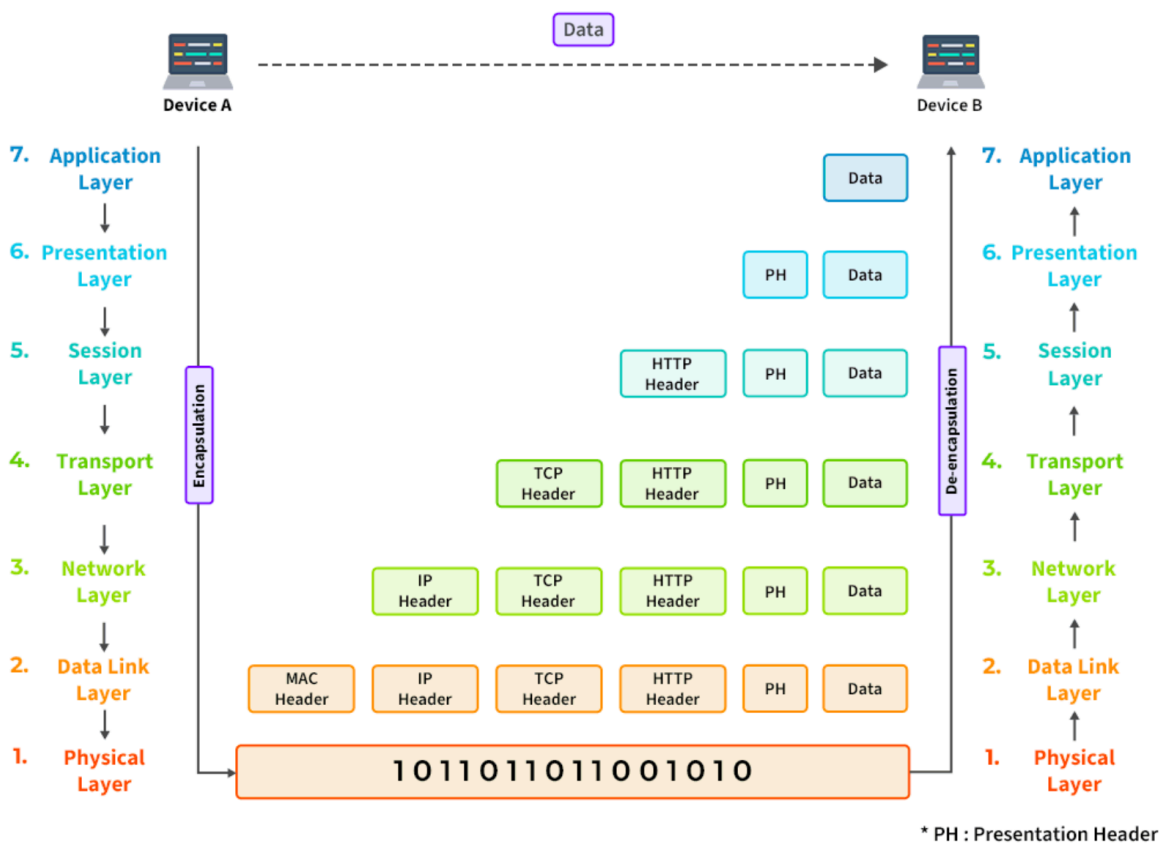




We can understand how data flows through OSI Model with the help of an example mentioned below. Let us suppose, Person A sends an e-mail to his friend Person B.

- ❖ Step 1: Person A interacts with e-mail application like Gmail, outlook, etc. Writes his email to send. (This happens at Application Layer).
- ❖ Step 2: At Presentation Layer, Mail application prepares for data transmission like encrypting data and formatting it for transmission.
- ❖ Step 3: At Session Layer, there is a connection established between the sender and receiver on the internet.
- ❖ Step 4: At Transport Layer, Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information.
- ❖ Step 5: At Network Layer, addressing of packets is done in order to find the best route for transfer.
- ❖ Step 6: At Data Link Layer, data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection.
- ❖ Step 7: At Physical Layer, Frames are transmitted in the form of electrical/optical signals over a physical network medium like ethernet cable or WiFi.

After the email reaches the receiver i.e. Person B, the process will reverse and decrypt the e-mail content. At last, the email will be shown on Person B email client.



Protocols Used in the OSI Layers

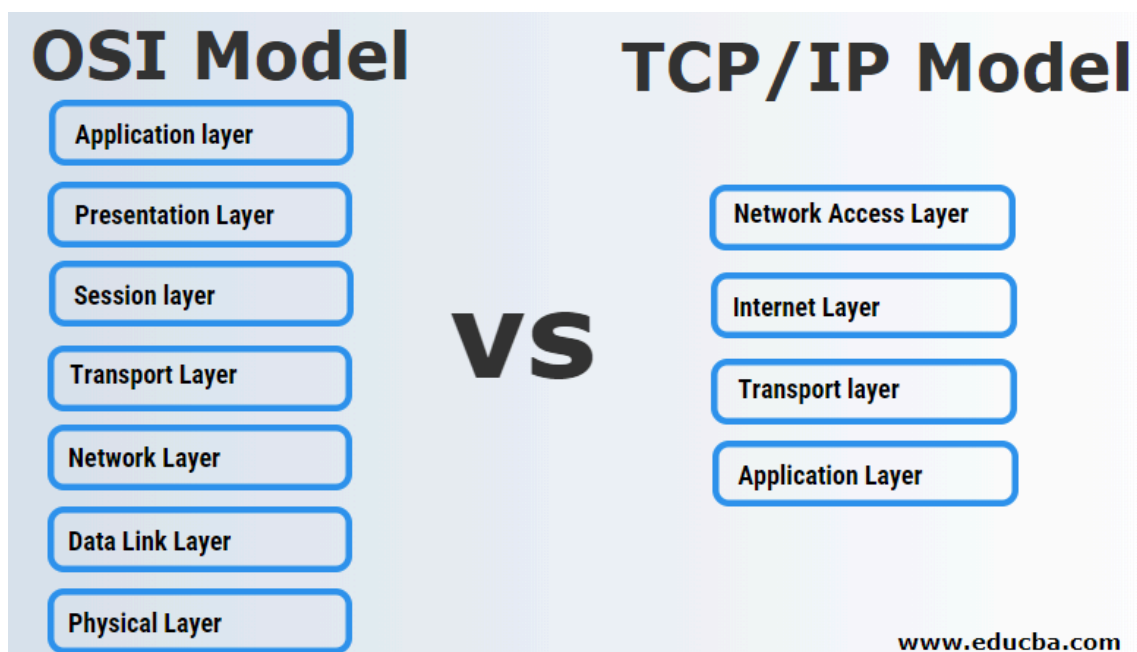
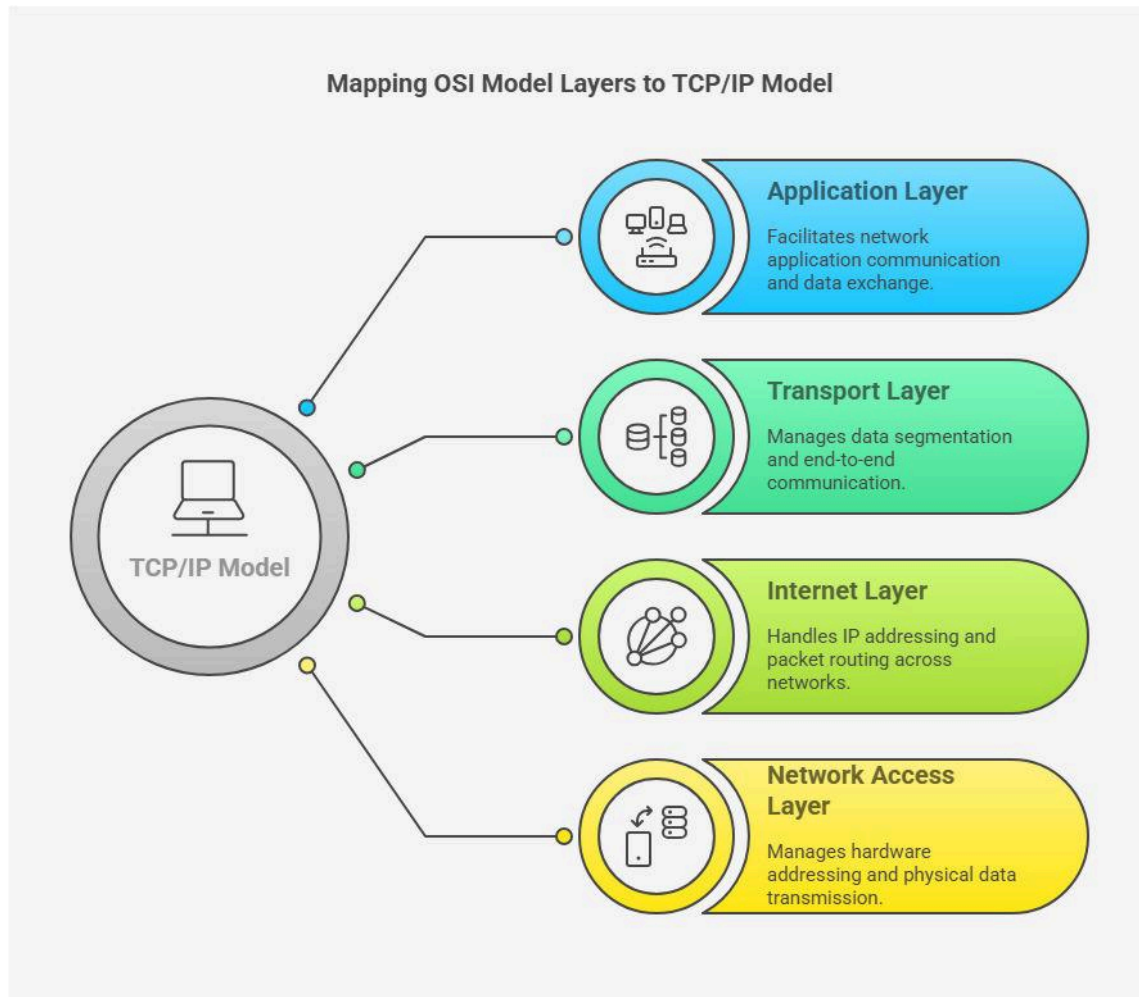
Layer	Working	Protocol Data Unit	Protocols
Physical Layer	Establishing Physical Connections between Devices.	Bits	USB, SONET/SDH, etc.
DataLink Layer	Node to Node Delivery of Message.	Frames	Ethernet, PPP, etc.
Network Layer	Transmission of data from one host to another, located in different networks.	Packets	IP, ICMP, IGMP, OSPF, etc.
Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segments (for TCP) or Datagrams (for UDP)	TCP, UDP, SCTP, etc.
Session Layer	Establishes Connection, Maintenance, Ensures Authentication and Ensures security.	Data	NetBIOS, RPC, PPTP, etc.
Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Data	TLS/SSL, MIME, etc.
Application Layer	Helps in identifying the client and synchronizing communication.	Data	FTP, SMTP, DNS, DHCP, etc.

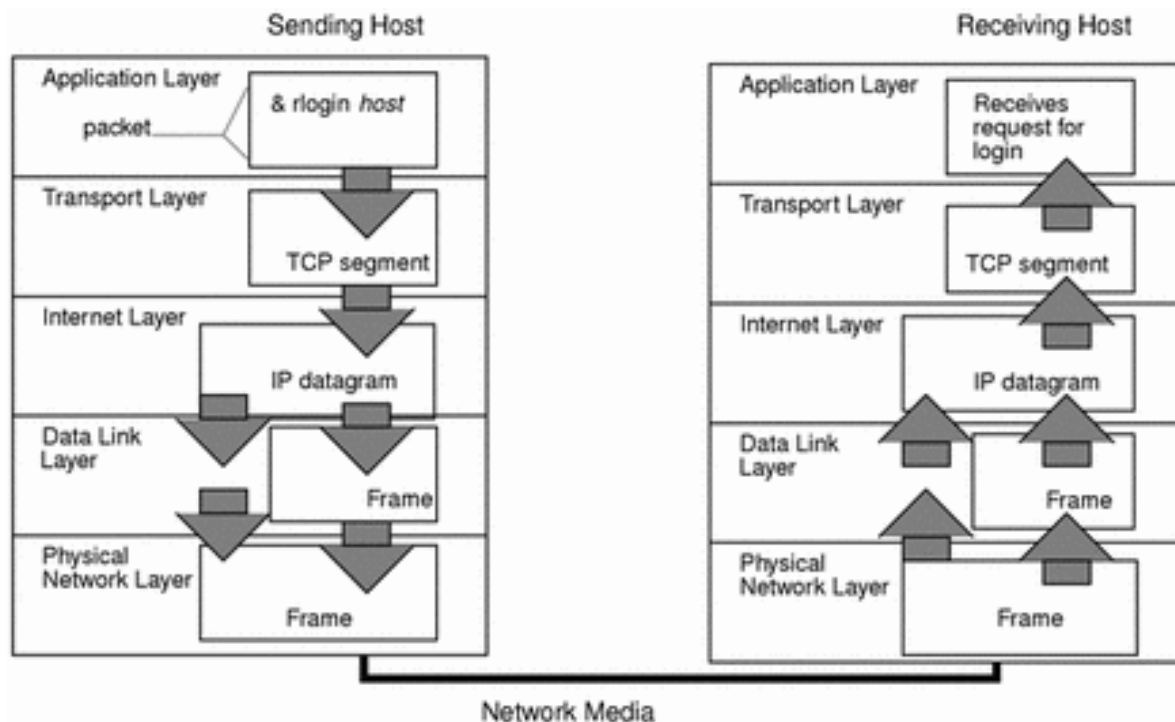
Why Does the OSI Model Matter

The OSI Model matters because it provides the user a clear structure of "how the data moves in the network?". As the OSI Model consists of 7 layers, each layer has its specific role, and due to which it helps in understanding, identifying and solving the complex network problems easily by focusing on one of the layers not the entire network.

***Note:** As the modern Internet does not prefer the OSI Model, but still, the OSI Model is still very helpful for solving network problems. It helps people understanding network concepts very easily.*

2.2 TCP/IP Model





The **TCP/IP model** (Transmission Control Protocol / Internet Protocol) is a **4-layer networking model** used to describe how data travels across the internet. It is the **practical model** used in real-world networking.

✚ Layers of TCP/IP Model

① Application Layer

👉 Top layer – interacts with users and applications

📌 Functions:

- Provides network services to applications
- Handles data formatting, encryption, session management

📌 Protocols:

- HTTP / HTTPS (web browsing)
- FTP (file transfer)
- SMTP, POP3, IMAP (email)
- DNS (domain name system)

🧠 Example:

When you open a website → HTTP is used here.

2 Transport Layer

👉 Responsible for **end-to-end communication**

📌 Functions:

- Data delivery (reliable/unreliable)
- Error control
- Flow control
- Segmentation (breaking data into parts)

📌 Protocols:

- TCP (reliable, connection-oriented)
- UDP (fast, connectionless)

🧠 Example:

- TCP → WhatsApp messages (reliable)
- UDP → Live streaming (fast)

3 Internet Layer

👉 Handles **logical addressing & routing**

📌 Functions:

- Assigns IP addresses
- Determines best path (routing)
- Packet forwarding

📌 Protocols:

- IP (IPv4 / IPv6)
- ICMP (error reporting)
- ARP (address resolution)

🧠 Example:

Like Google Maps → finds path from source to destination.

4 Network Access Layer (Link Layer)

👉 Deals with **physical transmission**

📌 Functions:

- Sends data over physical network
- Handles hardware addressing (MAC)
- Error detection

Technologies:

- Ethernet
- Wi-Fi

Example:

Actual cables, routers, switches sending data.

Data Flow (Encapsulation Process)

1. Application Layer → Data created
2. Transport Layer → Adds TCP/UDP header (segment)
3. Internet Layer → Adds IP header (packet)
4. Network Layer → Converts to frame & sends

 Reverse process at receiver → **Decapsulation**

TCP/IP vs OSI Model (Important)

Feature	TCP/IP	OSI
Layers	4	7
Usage	Practical (real-world)	Conceptual
Developed by	DARPA	ISO

Key Points for Exams

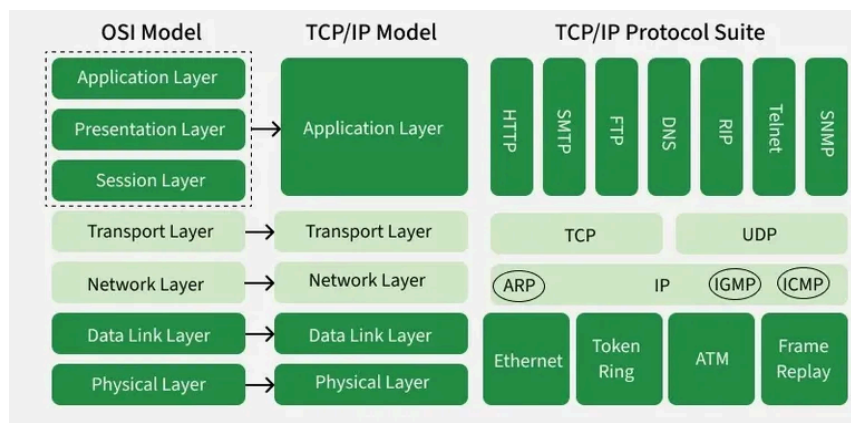
- TCP/IP has **4 layers**
- It is **used in the Internet**
- Combines multiple OSI layers
- Focuses on **communication, routing, and delivery**

One-Line Answer

 *TCP/IP is a 4-layer networking model that defines how data is transmitted over the internet using protocols like TCP and IP.*

2.3 OSI vs TCP/IP Model (Comparison Table)

Feature	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Number of Layers	7 Layers	4 Layers
Layers Names	Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
Developed By	ISO (International Organization for Standardization)	DARPA (US Department of Defense)
Nature	Theoretical model	Practical model
Usage	Used for understanding & teaching	Used in real-world networking (Internet)
Layer Separation	Strict separation of layers	Less strict (layers combined)
Session & Presentation Layers	Present (separate layers)	Not present (merged into Application layer)
Protocol Dependency	Protocol independent	Protocol dependent
Data Flow Approach	Vertical approach	Horizontal approach
Flexibility	Less flexible	More flexible



3. Physical Layer

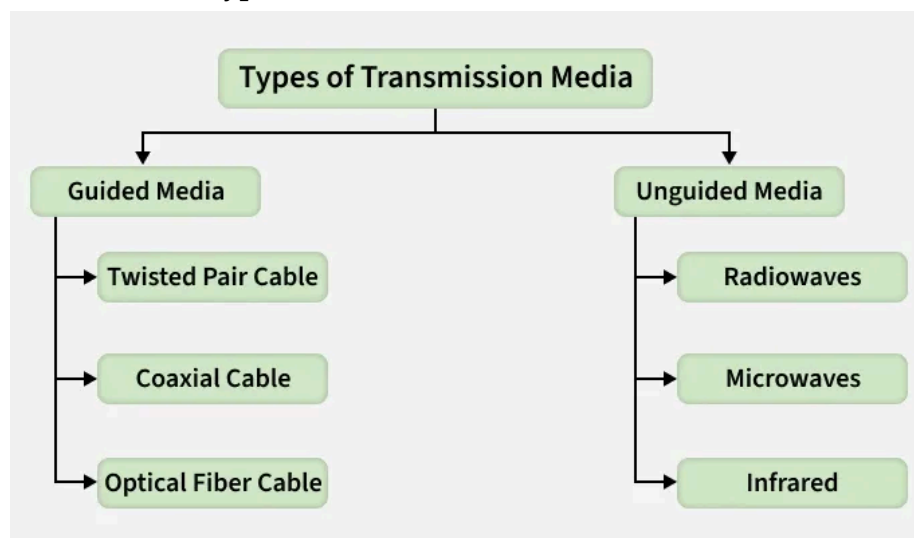
3.1 Transmission Media in Computer Networks(guided/Unguided)

Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another within a computer network. It forms the fundamental pathway through which information is transmitted, ensuring connectivity between networked devices.

The selection of a transmission medium depends on factors such as:

- Transmission distance
- Data transfer speed (bandwidth)
- Susceptibility to interference and noise
- Cost and installation requirements

Based on the nature of the transmission path, transmission media are broadly classified into two main types:



Guided Media

Guided Media also known as wired or bounded transmission media, refers to transmission media in which data signals are transmitted through a physical path using cables. The signal is confined and guided along a fixed route, providing controlled communication between network devices.

- Uses physical cables to transmit data signals.
- Provides dedicated and well-defined transmission paths.
- Major types of guided media included Twisted Pair Cables, Coaxial Cables and Optical Fiber Cables.
- Offers higher data transmission rates compared to most wireless media.

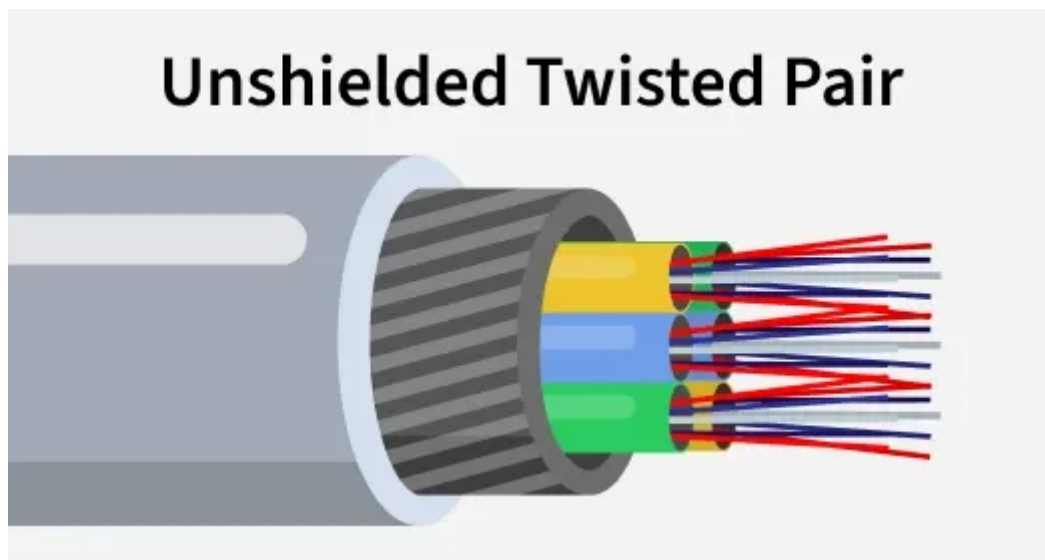
- Provides better security due to physical connectivity and limited signal leakage.
- Suitable for short to long-distance communication, depending on the cable type used.

1. Twisted Pair Cable

A twisted pair cable consists of two individually insulated copper conductors twisted together in a helical pattern to minimize electromagnetic interference (EMI) and crosstalk between adjacent pairs. Multiple twisted pairs are typically enclosed within a protective outer sheath, making this cable one of the most commonly used and cost-effective guided transmission media.

- Twisting of conductors helps reduce interference and signal degradation.
- Multiple pairs can be bundled together within a single cable.
- Widely used due to low cost, flexibility, and ease of installation.
- Commonly used in telephone systems and local area networks (LANs).

(a) **Unshielded Twisted Pair (UTP):** UTP cable consists of twisted copper wire pairs without any additional metallic shielding. The twisting provides basic protection against interference, making UTP suitable for environments with moderate noise levels.



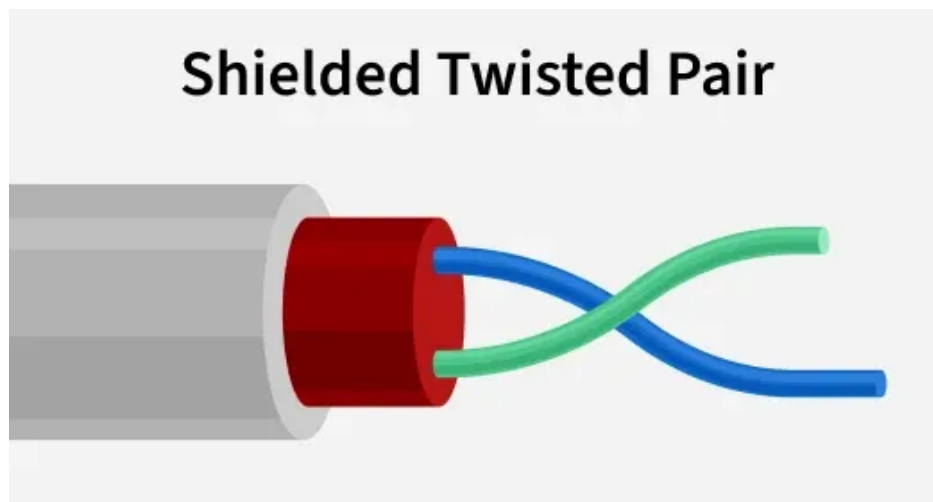
Advantages:

- Least expensive among guided transmission media.
- Easy to install and maintain due to its lightweight and flexible nature.
- Supports high data transmission rates over short distances.
- Widely used in Ethernet-based LANs and telephone networks.

Disadvantages:

- Lower performance and capacity compared to Shielded Twisted Pair (STP).
- Limited transmission distance due to signal attenuation.
- More susceptible to noise and electromagnetic interference.

(b) Shielded Twisted Pair (STP): Shielded Twisted Pair (STP) cable consists of twisted pairs of insulated copper conductors enclosed within an additional metallic shielding layer, such as a foil or braided shield. This shielding provides enhanced protection against external electromagnetic interference (EMI) and reduces crosstalk, resulting in more stable and reliable data transmission.



Advantages

- Provides better performance at higher data transmission rates compared to UTP.
- Significantly reduces electromagnetic interference and crosstalk.
- Ensures more reliable and stable communication, especially in noisy environments.
- Suitable for high-speed Ethernet networks and voice and data transmission systems.

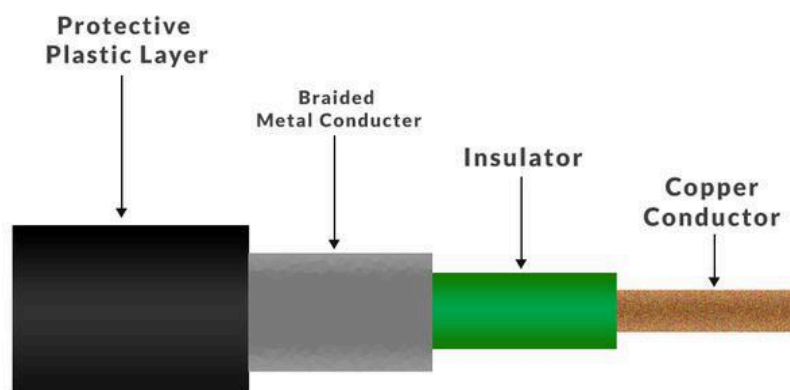
Disadvantages

- Higher cost compared to Unshielded Twisted Pair (UTP).
- More complex installation and termination due to shielding requirements.
- Bulkier and less flexible, making cable management more difficult.

2. Coaxial Cable

Coaxial Cable consists of a central copper conductor surrounded by a dielectric insulating layer, a metallic shielding layer, and an outer protective jacket. This structure allows signals to be transmitted with better protection against noise and interference compared to twisted pair cables.

- Supports data transmission in both baseband mode (single channel) and broadband mode (multiple channels).
- Provides higher bandwidth than twisted pair cables.
- Strong shielding makes it more resistant to noise, crosstalk, and electromagnetic interference (EMI).
- Commonly used in cable television (CATV), broadband networks, and analog television systems.
- More durable and reliable due to its layered construction.
- Easier to install and maintain compared to optical fiber cables.



Advantages:

- Supports higher bandwidth than twisted pair cables.
- Offers better signal quality and reliability.
- Allows multiple channels using frequency division multiplexing (FDM).
- Less affected by external electromagnetic interference.

Disadvantages:

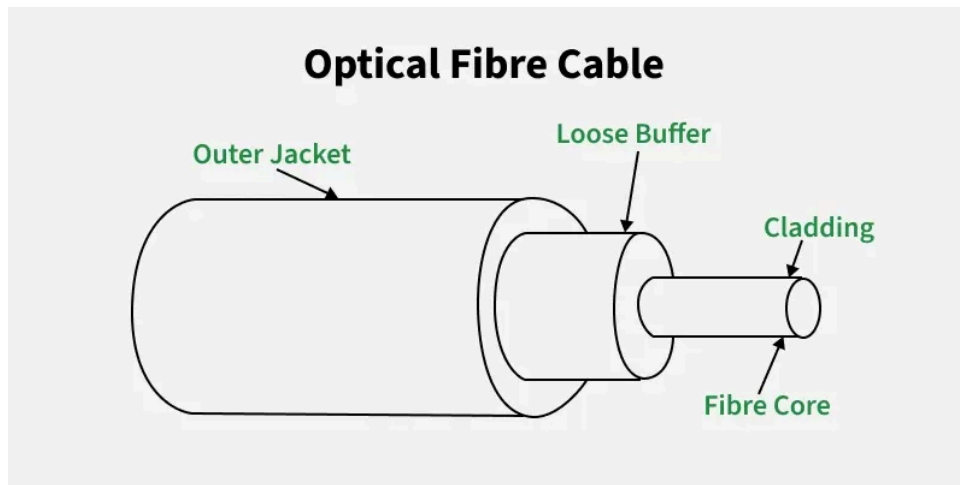
- More expensive than twisted pair cables.
- Requires proper grounding to ensure safety and minimize interference.
- Bulkier and less flexible due to multiple layers.
- More vulnerable to security breaches, as the cable can be physically tapped.

3. Optical Fiber Cable

Optical Fiber Cable is a guided transmission medium that transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection. The core is surrounded by a cladding layer with a lower refractive index, which confines the light within the core and enables high-speed, long-distance data transmission.

- Supports very large data volumes at extremely high speeds.
- Can operate in unidirectional or bidirectional communication modes.

- WDM (Wavelength Division Multiplexer) allows multiple light signals to be transmitted simultaneously over a single fiber.
- Widely used where high bandwidth, long distance, and low signal loss are required.



Advantages:

- Provides very high bandwidth and data-carrying capacity.
- Lightweight and compact, making installation easier over long distances.
- Exhibits low signal attenuation, suitable for long-distance communication.
- Immune to electromagnetic interference (EMI) and electrical noise.
- Resistant to corrosion and harsh environmental conditions.

Disadvantages:

- Complex installation and maintenance due to precise splicing and handling requirements.
- High initial installation and equipment cost.
- Fragile compared to copper cables and requires careful handling.

Applications:

- ☐ **Medical:** Used in endoscopy, imaging systems, and laser-based surgical procedures.
- ☐ **Defense and Aerospace:** Supports secure, high-speed, and interference-free communication.
- ☐ **Communication:** Widely used in internet backbone networks, undersea cables, and telecommunication systems.
- ☐ **Industrial and Automotive:** Used for sensing, lighting, data communication, and safety systems.

Unguided Media

Unguided media, also known as wireless or Unbounded transmission media, uses electromagnetic waves to transmit data without any physical medium. Signals

propagate through free space such as air or vacuum. The main types of unguided media are radio waves, microwaves, and infrared waves.

Features:

- Signals propagate through air or free space
- Less secure due to broadcast nature
- Suitable for long-distance communication

1. Radio Waves

Radio waves are a type of electromagnetic wave that can easily be generated and can propagate through buildings and other obstacles. They do not require line-of-sight between transmitting and receiving antennas, making them highly suitable for broadcast communication and wireless data transmission.

Frequency Range:

3 kHz – 300 GHz

Applications:

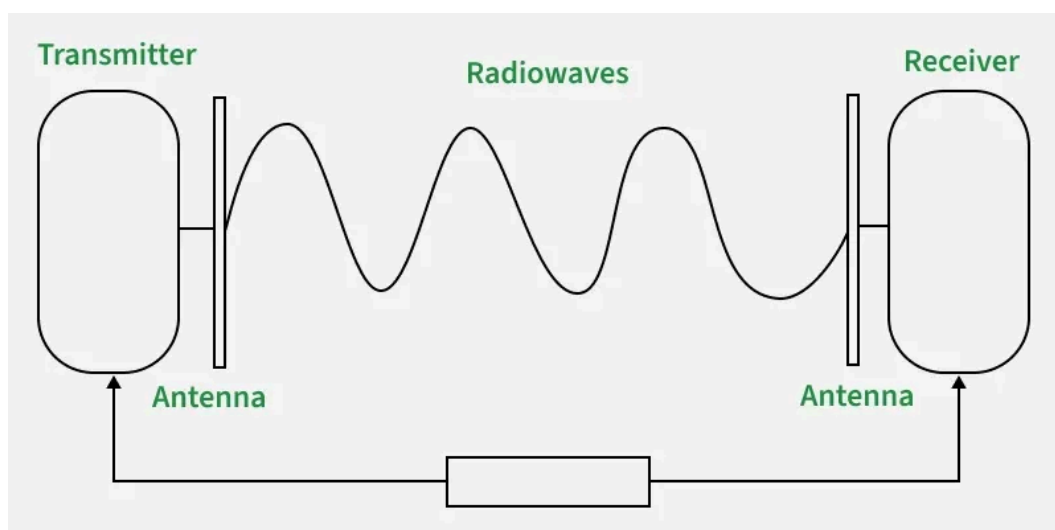
- AM and FM radio broadcasting, television transmission, cordless phones, and wireless communication.

Types of Radio Waves:

- **Shortwave:** AM radio broadcasting
- **VHF (Very High Frequency):** FM radio and television
- **UHF (Ultra High Frequency):** Television and mobile communication

Components:

- **Transmitter:** Generates and modulates the signal
- **Receiver:** Receives and demodulates the signal



2. Microwaves

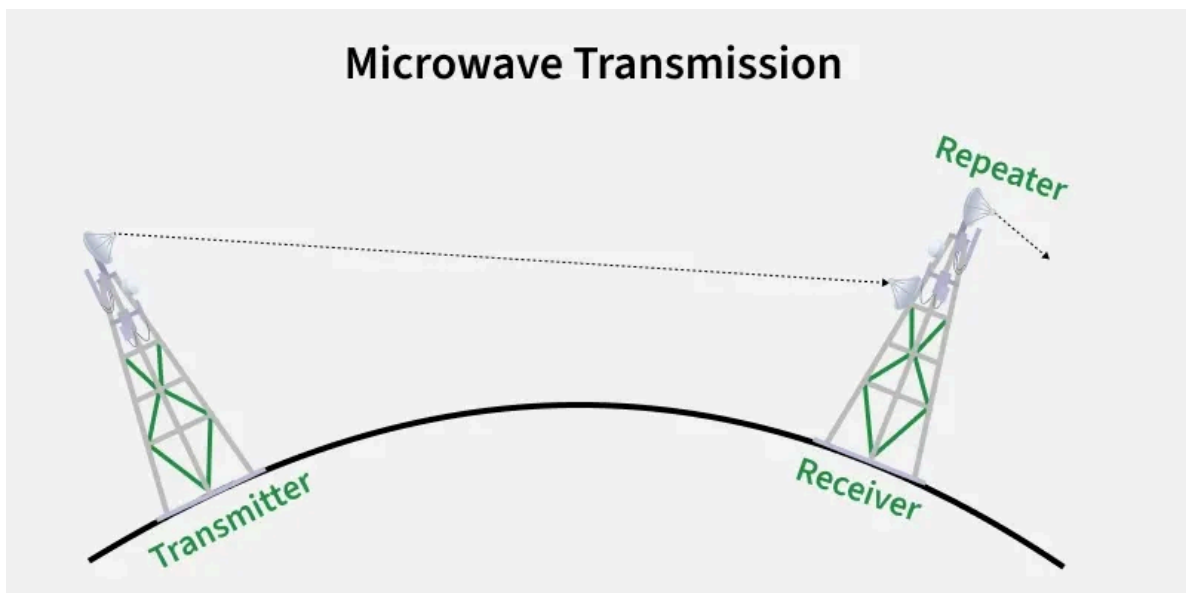
Micro waves are a form of unguided transmission media that use line-of-sight communication, where the transmitting and receiving antennas must be properly aligned. The transmission range depends on the height of the antennas. Microwaves operate in the frequency range of 1 GHz to 300 GHz and are widely used in mobile communication, satellite links, and television distribution.

Advantages:

- Cost-effective compared to laying physical cables
- No need for land acquisition
- Suitable for communication over difficult terrains and oceans
- Supports high data transmission rates

Disadvantages:

- Less secure without proper encryption
- Susceptible to weather conditions such as rain and fog
- Affected by obstacles due to line-of-sight requirement
- High cost of antenna design, installation, and maintenance



3. Infrared

Infrared waves are used for short-range wireless communication and operate in the frequency range of 300 GHz to 400 THz. They cannot penetrate solid obstacles, which limits their range but helps minimize interference between nearby systems. Infrared waves are commonly used in TV remote controls, wireless keyboards, mice, and printers.



Television



Infrared Radiations

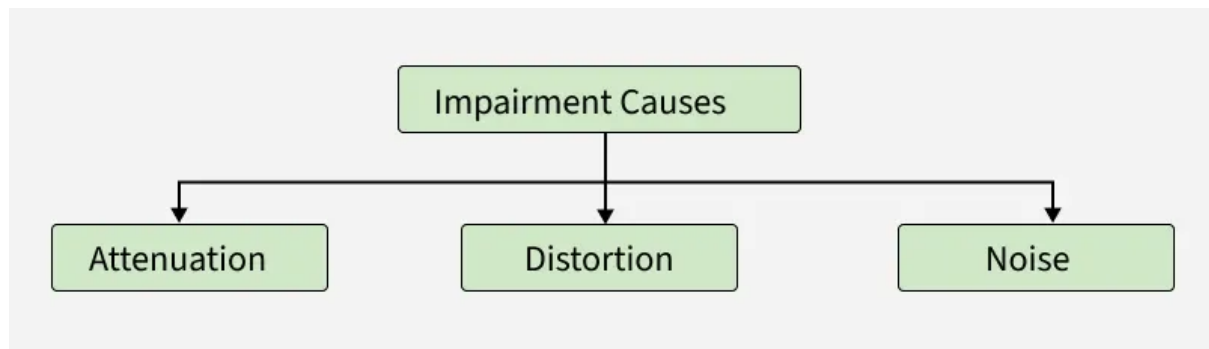


Remote

Feature	Radio Waves	Microwaves	Infrared Waves
Direction	Omnidirectional in nature	Highly directional	Directional, requires line of sight (LOS)
Penetration	Can penetrate buildings and obstacles easily due to low frequency	Poor penetration; line of sight required	Cannot penetrate obstacles
Frequency Range	3 kHz – 300 GHz	1 GHz – 300 GHz	300 GHz – 400 THz
Security	Low security	Medium security (encryption possible)	Relatively high security but limited range
Attenuation	Moderate attenuation	Variable attenuation, depends on weather conditions	High attenuation
Licensing	Some frequency bands require government licensing	Most frequency bands are licensed by the government	Generally unlicensed
Cost	Low to moderate cost	High cost	Low cost
Distance	Supports long-distance communication	Supports long-distance communication	Suitable only for very short-distance communication

Causes of Transmission Impairment

Transmission impairment refers to the loss or distortion of signals during data transmission, leading to errors or reduced quality in communication. Common causes include signal distortion, attenuation, and noise all of which can affect the clarity and reliability of transmitted data



- **Attenuation:** Loss of signal strength as the signal propagates over a transmission medium due to resistance and energy loss. In analog systems, amplifiers are used to increase signal strength, while in digital communication, repeaters or regenerators are used to restore the original signal.
- **Distortion:** Occurs when the shape of the transmitted signal changes. This typically happens in composite signals where different frequency components travel at different speeds, causing phase shifts and timing differences at the receiver.
- **Noise:** It refers to unwanted electrical or electromagnetic signals that interfere with the original signal, potentially corrupting data. Common types of noise include thermal noise, induced noise, crosstalk, impulse noise.

Factors Considered for Designing the Transmission Media

- **Bandwidth:** It refers to the range of frequencies that a transmission medium can support. Higher bandwidth allows higher data transmission rates, assuming other factors such as noise and attenuation remain constant.
- **Transmission Impairment :** Transmission Impairment occurs when the received signal differs from the transmitted signal. Signal quality will be impacted as a result of transmission impairment.
- **Interference:** It is the disturbance of a signal caused by the addition of unwanted signals from external sources such as electromagnetic radiation or neighboring channels. It degrades signal clarity and may result in data corruption.

Transmission Media	Application
Unshielded Twisted Pair (UTP)	Local Area Networks (LAN), telephones
Shielded Twisted Pair (STP)	Industrial networks, environments with high interference
Optical Fiber Cable	Long-distance communication, internet backbones
Coaxial Cable	Cable TV, broadband internet, CCTV
Radio	Wireless communication, AM/FM radio, mobile phones
Infrared	Remote controls, short-range communication
Microwave	Satellite communication, radar, long-distance links

Planar Transmission Lines

Planar transmission lines are specialized guided structures used to carry high-frequency electrical signals over very short distances within electronic circuits and printed circuit boards. Although they are not used as network communication media, they apply the same fundamental transmission principles used in guided transmission media, such as controlled signal propagation, impedance matching, and interference reduction.

- They follow the same physical layer concepts as guided transmission media.
- Used where very high-frequency signals must be transmitted with minimal loss.
- Help demonstrate how transmission media principles are applied inside electronic hardware.
- Commonly studied in advanced physical-layer and RF design contexts.

Stripline

Stripline is a planar guided transmission structure in which a conducting strip is embedded between two parallel ground planes, allowing high-frequency signals to propagate with excellent shielding and minimal electromagnetic interference.

- Fully enclosed structure provides high EMI immunity
- Supports TEM mode propagation
- Offers better signal isolation and stability
- Used in microwave and high-frequency circuit design
- Suitable for high-performance RF systems

Microstripline

Microstripline is a planar transmission structure consisting of a conducting strip placed on a dielectric substrate with a ground plane on the opposite side, making it widely used for high-frequency signal transmission in printed circuit boards.

- Simple and cost-effective PCB implementation
- Partially exposed fields → lower shielding than stripline
- Supports quasi-TEM mode propagation
- Common in RF circuits, antennas, and microwave PCBs
- Preferred for compact electronic designs

3.2 Signals (Analog/Digital)

Difference Between Analog and Digital signal

Analog and digital signals are two fundamental types of electrical signals used to transmit information in various electronic systems. Analog signals are continuous and vary smoothly over time, while digital signals are discrete and represented by a series of discrete values.

Understanding the difference between these signals is important in telecommunications, electronics, and signal processing. In this article, we will go through the difference between analog and digital signals.

Table of Content

1. What is an Analog Signal?
2. What is Digital Signal?
3. Difference Between Analog And Digital Signal

What is an Analog Signal?

Analog signal is a type of signal that represents continuous data using a continuous range of values. In other words, it can take on any value within a certain range. Analog signals are characterized by their smooth and continuous nature

For example, imagine a sound wave. In analog form, the sound wave is represented by a continuously varying electrical signal that mirrors the fluctuations in air pressure caused by the sound.

What is Digital Signal?

Digital signal is a type of signal that represents data as a sequence of discrete values, typically using binary numbers, it also contain different voltage values. Unlike analog signals, which vary continuously over time, digital signals are discrete and quantized, meaning they only take on specific, distinct values.

In digital communication and computing systems, information is encoded into digital signals for transmission, processing, and storage. These signals are commonly used in telecommunications, audio and video processing, computer networks, and many other applications.

Difference Between Analog And Digital Signal

The difference between analog signal and digital signal could be understood from the table given below:

Basis	Analog Signal	Digital Signal
Definition	Analog signals represent continuous variations in magnitude over time.	Digital signals are discrete and quantized, with specific values.
Signal Type	Continuous waveforms	Discrete signals
Processing	Requires complex processing for manipulation	Easier to process and manipulate digitally
Storage	Less efficient for storage due to continuous nature	More efficient for storage due to discrete values
Bandwidth	Typically requires more bandwidth	Requires less bandwidth for transmission
Examples	Analog audio signals, analog radio waves, human voice, etc.	Digital audio signals, digital data streams, computers, etc.
Errors	Susceptible to noise and distortion	More resistant to noise and distortion
Circuit Component	Amplifiers, filters, continuous-wave oscillators	Microprocessors, binary counters, logic gates
Signal Values	Infinite range of values	Limited to discrete values
Conversion	No conversion required	Analog-to-digital conversion (ADC) required
Applications	Analog signals are used in electric fan, landlines, radio frequency communications, etc.	Digital signals are used in computers, smartphones, digital sensors, digital imaging, etc.

3.3 Bandwidth and data rate

Difference between Bandwidth and Data Rate

In computer network bandwidth and data rate both are important concepts to understand network performance. Although they are very closely related, but they have distinct differences. In this article we will see the difference between bandwidth and data rate in detail.

What is the Bandwidth?

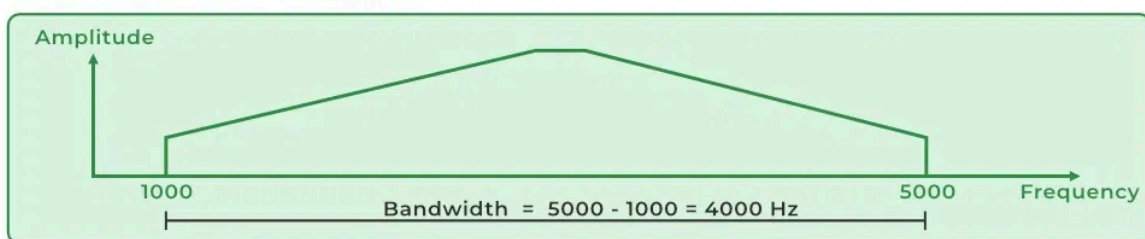
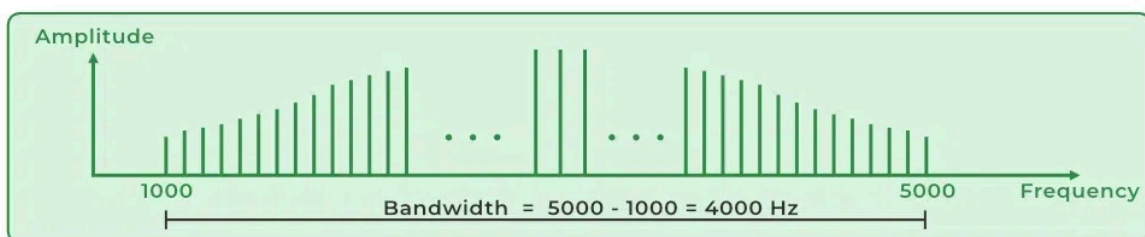
Bandwidth is defined as the width of the spectrum. Spectrum is the range of frequencies contained in the signal. Since bandwidth is the width of the spectrum and spectrum is a range of frequencies, bandwidth is measured in Hertz or Mhz.

Advantages of Bandwidth

- It has a Higher Data Transfer Capacity which allows more data to be transferred simultaneously.
- It can be used to improve performance with more bandwidth.
- It helps to enhance the Quality of Service as it supports high-quality video streaming, online gaming, etc.

Disadvantages of Bandwidth

- The Higher bandwidth connections are typically more expensive.
- In some cases, the full potential of high bandwidth can't be utilized.
- It needs more advanced infrastructure which can be expensive



What is the Data Rate?

Data Rate is defined as the amount of data transmitted during a specified period over a network. It is the speed at which data is transferred from one device to another or

between a peripheral device and the computer. It is generally measured in Megabits per second(Mbps).

Advantages of Data Rate

- Data rate helps to show the actual speed of data transfer.
- The Data Rate can help in optimizing network resources and managing traffic.
- A high data rate ensures a better user experience with less buffering.

Disadvantages of Data Rate

- The Data Rate can go with Variable Speeds.
- The Data Rate Depends upon the Bandwidth.
- Many External Factors can influence the data rate.

Difference between Bandwidth and Data Rate

Bandwidth	Data Rate
It is the potential of carrier channels that can carry data.	It is the amount of data transmitted during a specified period over a network.
It is the difference between the range of frequencies.	It is the speed of data transmission.
Normally it is measured in Hz or kHz or MHz.	It is normally measured in Mbps or MBps.
It refers to the maximum data transmission capacity of the channel.	It refers to the actual data transmission speed.
It is a physical layer property in the OSI model.	While it is common in all layers.
It shows the capacity of the channel.	It shows the present speed of data transmission.
It does not depend on the properties of the sender or receiver.	While it gets affected by the sender or receiver.

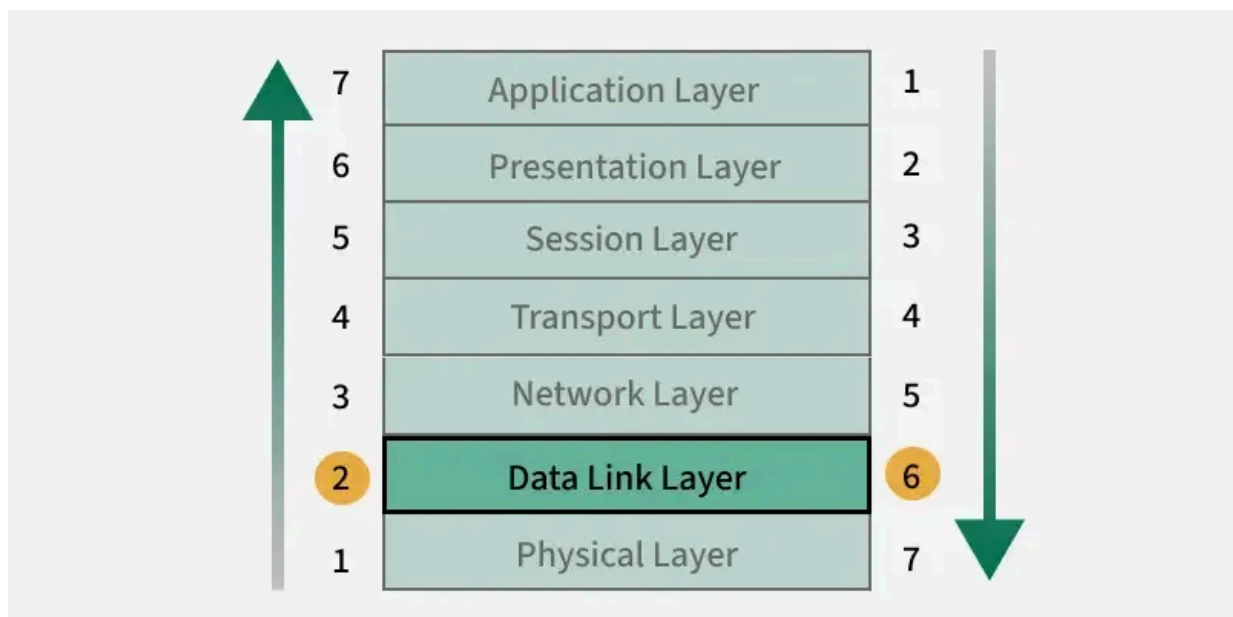
Conclusion

Both Bandwidth and Data Rate are closely connected. Both of them are necessary in the Data Transmission process which is very much needed for Computer Networking. Knowing the difference between them will help to figure out which element should be used for what purposes.

4.Data Link Layer in OSI Model

The data link layer is the second layer from the bottom in the OSI (Open System Interconnection) network architecture model.

- Responsible for the node-to-node delivery of data within the same local network.
- Major role is to ensure error-free transmission of information.
- Also responsible for encoding, decoding, and organizing the outgoing and incoming data.
- Considered as the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers.



Sub-Layers of The Data Link Layer

The data link layer is further divided into two sub-layers, which are as follows:

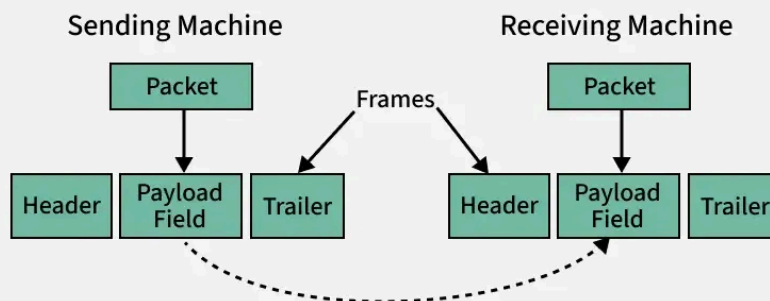
1. **Logical Link Control (LLC):** This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.
2. **Media Access Control (MAC):** MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

Functions of The Data-link Layer

Functions of Data Link Layer

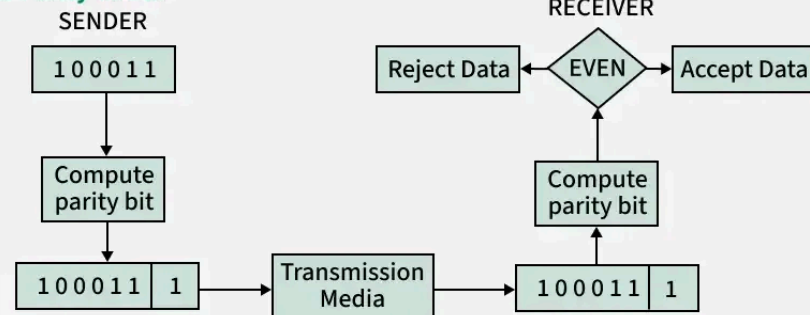
1. Framing
2. Error Detection - Parity Check
3. Error Detection - Checksum
4. Error Detection - CRC
5. Error Correction
6. Flow Control
7. Addressing

1. Framing



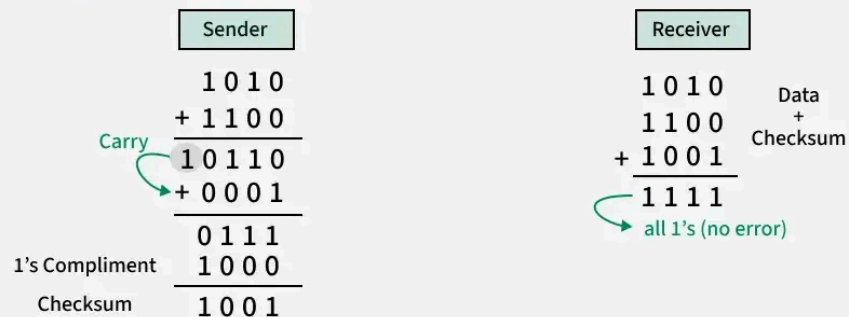
2. Error Detection

1. Parity Check



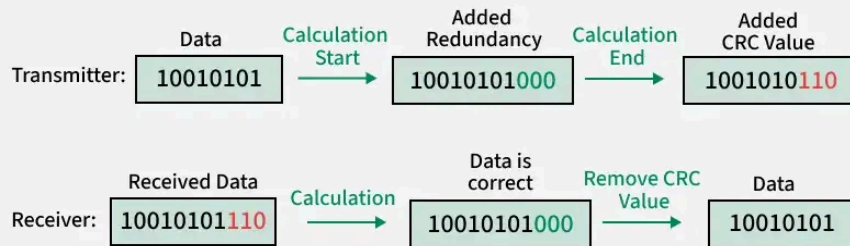
2. Error Detection

2. Checksum



2. Error Detection

3. Cyclic Redundancy Check



3. Error Correction

- **Automatic-repeat request (ARQ)**

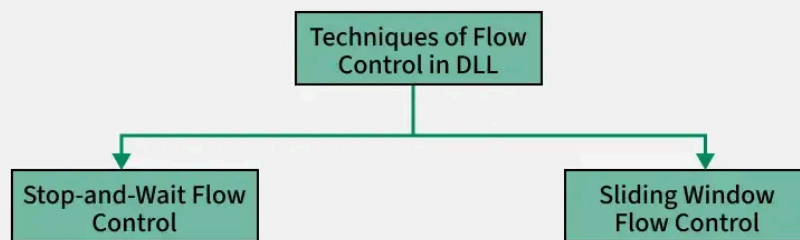
ARQ is a feedback-based method where the receiver requests the sender to retransmit the data if errors are detected.

- **Forward error correction (FEC):**

FEC is a method where the sender includes additional redundant data, allowing the receiver to detect and correct errors without retransmission.

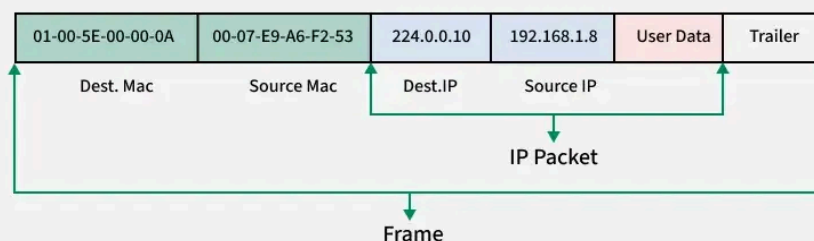
4. Flow Control

The Data Link Layer ensures flow control by synchronizing the sender's and receiver's speeds to prevent buffer overflow and frame loss.



5. Addressing

The Data Link Layer adds source and destination MAC addresses in the frame header for node-to-node delivery.



Read more about Data Link Layer Services.

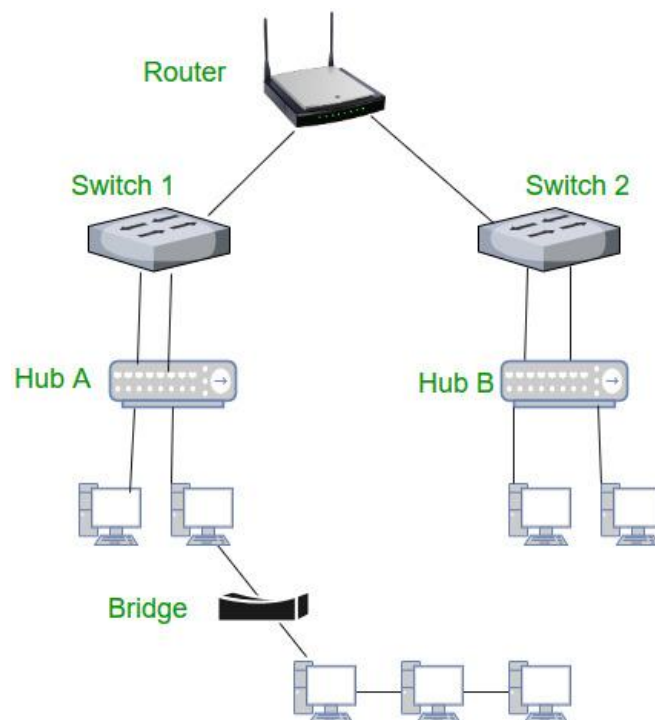
Protocols in Data link layer

There are various protocols in the data link layer, which are as follows:

- 1) Synchronous Data Link Protocol (SDLC)
- 2) High-Level Data Link Protocol (HDLC)
- 3) Serial Line Interface Protocol (SLIP)
- 4) Point to Point Protocol (PPP)
- 5) Link Access Procedure (LAP)
- 6) Link Control Protocol (LCP)
- 7) Network Control Protocol (NCP)

Devices Operating at the Data Link Layer

All these devices rely on MAC addresses for efficient frame delivery and play a crucial role in local network communication and access control.



1. Switch

- A switch is a key device in the Data Link Layer.
- It uses MAC addresses to forward data frames to the correct device within a network.
- Works in local area networks (LANs) to connect multiple devices.

2. Bridge

- A bridge connects two or more LANs, creating a single, unified network.

- Operates at the Data Link Layer by forwarding frames based on MAC addresses.
- Used to reduce network traffic and segment a network.

3. Network Interface Card (NIC)

- A NIC is a hardware component in devices like computers and printers.
- Responsible for adding the MAC address to frames and ensuring proper communication with the network.
- Operates at the Data Link Layer by preparing and sending frames over the physical medium.

4. Wireless Access Point (WAP)

- A WAP allows wireless devices to connect to a wired network.
- Operates at the Data Link Layer by managing wireless MAC addresses.
- Uses protocols like Wi-Fi (IEEE 802.11) to communicate with devices.

5. Layer 2 Switches

- These are specialized switches that only operate at Layer 2, unlike multi-layer switches.
- Responsible for frame forwarding using MAC address tables.

Note: The Data Link Layer can be targeted by attacks like MAC spoofing or ARP poisoning. Understanding how devices and frames operate at this layer helps detect and mitigate such threats.

Limitations of Data Link Layer

- ❖ Limited Scope: It operates only within a local network and cannot handle end-to-end communication across different networks.
- ❖ Increased Overhead: Adding headers, trailers, and redundant data (for error correction) increases the size of transmitted data.
- ❖ Error Handling Dependency: While it can detect and correct some errors, it relies on upper layers for handling more complex issues.
- ❖ No Routing Capability: The Data Link Layer cannot make routing decisions. It only ensures delivery within the same network segment.
- ❖ Resource Usage: Flow control and error correction mechanisms may consume extra processing power and memory

Applications of Data Link Layer

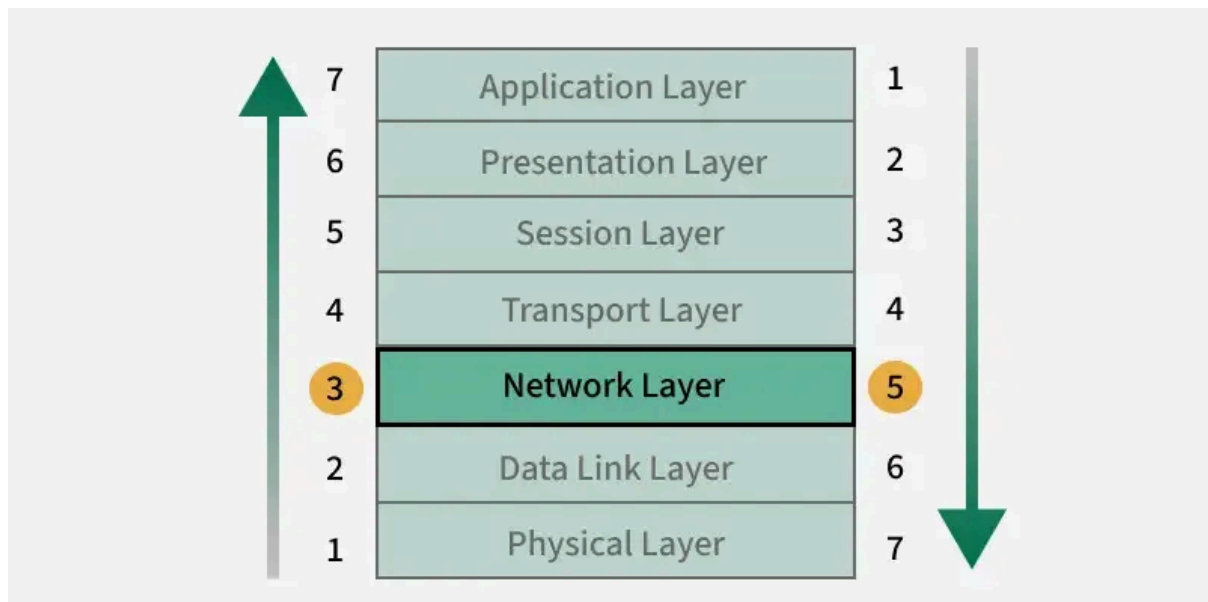
- Local Area Networks (LANs): Enables reliable communication between devices within a local network using protocols like Ethernet (IEEE 802.3).
- Wireless Networks (Wi-Fi): Manages communication between devices in wireless networks via protocols like IEEE 802.11 hence, handling media access and error control.

- **Switches and MAC Addressing:** Facilitates the operation of switches by using MAC addresses to forward data frames to the correct device within the network.
- **Point-to-Point Connections:** Used in protocols like PPP (Point-to-Point Protocol) to establish and manage direct communication between two nodes.

5. Network Layer in OSI Model

The Network Layer is the third layer of the OSI model responsible for logical addressing, routing, and end-to-end packet delivery across interconnected networks.

- Handles logical (IP) addressing of devices
- Determines optimal routing paths between networks
- Enables host-to-host communication across multiple networks



Note: Unlike the Data Link Layer, which focuses only on node-to-node delivery within a single network segment, the Network Layer ensures that data travels from the source host to the destination host, even if they are located on different networks.

Key Responsibilities of the Network Layer

Some of the key responsibilities of network layer are:

- **Logical Addressing:** Assigns unique IP addresses to devices, ensuring accurate identification and communication across networks.
- **Packetization:** Encapsulates transport layer segments into packets for efficient transmission.
- **Host-to-Host Delivery:** Ensures reliable delivery of packets from the sender to the intended receiver across diverse networks.
- **Forwarding:** Moves packets from the input interface of a router to the appropriate output interface based on the destination IP.

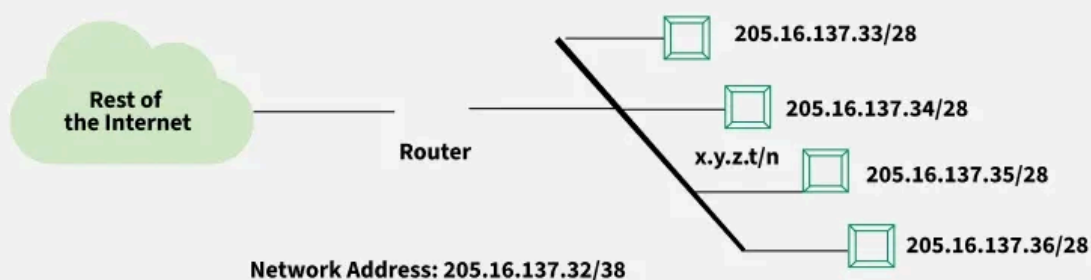
- **Routing:** Determines the optimal path for packets to travel across multiple networks using routing algorithms and protocols.
- **Fragmentation and Reassembly:** Splits large packets into smaller fragments to match the maximum transmission unit (MTU) of a network, and reassembles them at the destination.
- **Subnetting:** Divides larger networks into smaller subnetworks for efficient addressing and traffic management.
- **Network Address Translation (NAT):** Maps private IPs to public IPs for internet communication, conserving address space and adding security.

Functions of Network Layer

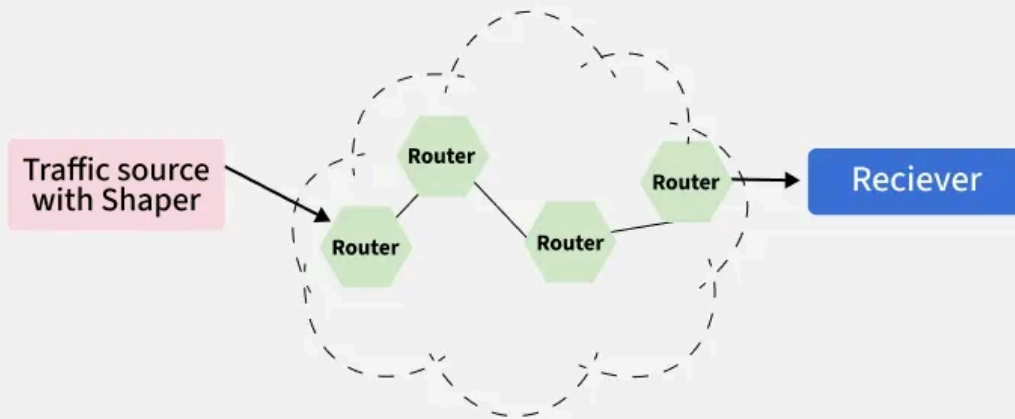
- | | |
|------------------------------|--|
| 1. Assigning Logical Address | 5. Fragmentation and Reassembly of packets |
| 2. Packetizing | 6. Logical Subnetting |
| 3. Host-to-host delivery | 7. Network Address Translation |
| 4. Forwarding | 8. Routing |

1. Assigning Logical Address

It assigns unique IP addresses to devices for identification across networks.

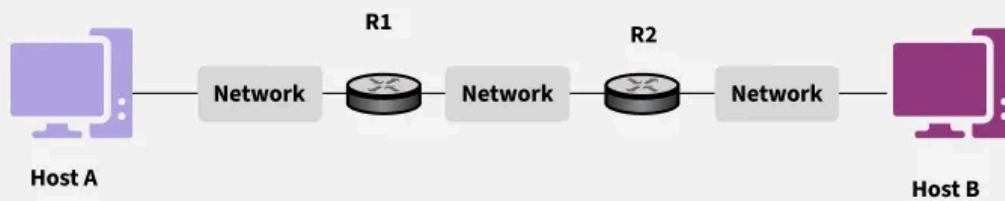


2. Packetizing



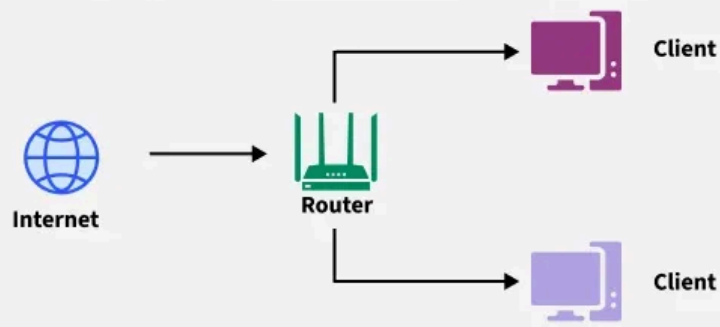
3. Host-to-host delivery

It ensures that data is reliably transmitted between two hosts across a network

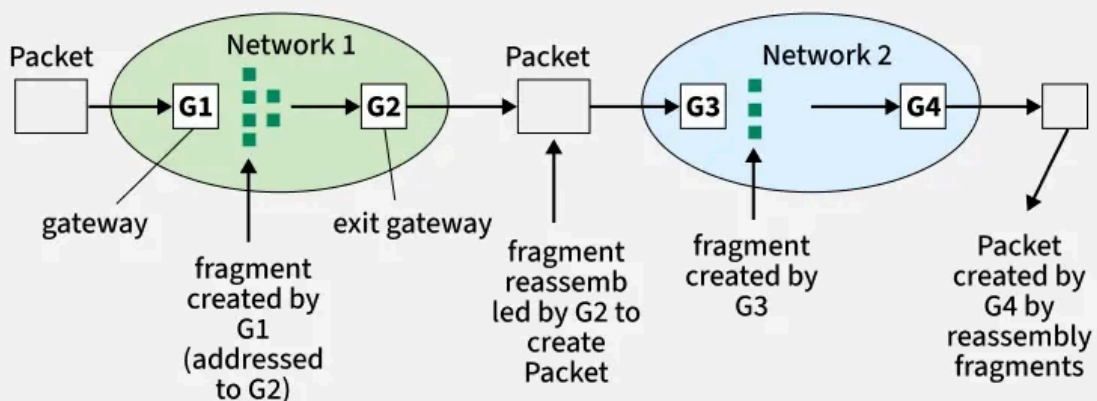


4. Forwarding

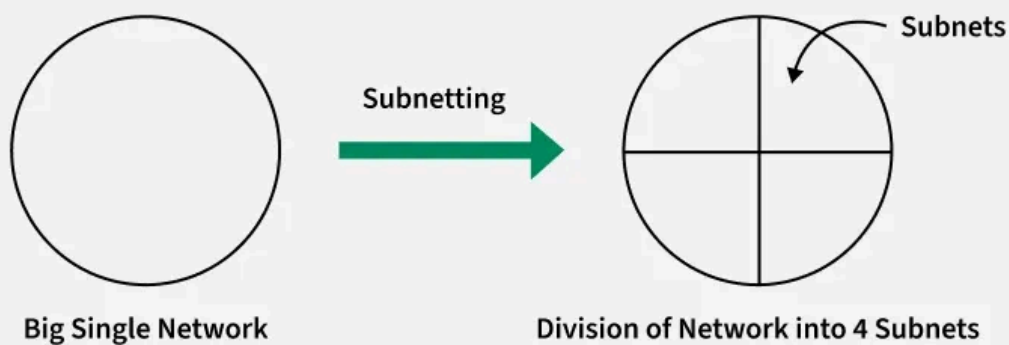
Forwarding moves packets through a router to the correct outgoing interface based on their destination.



5. Fragmentation and Reassembly of packets

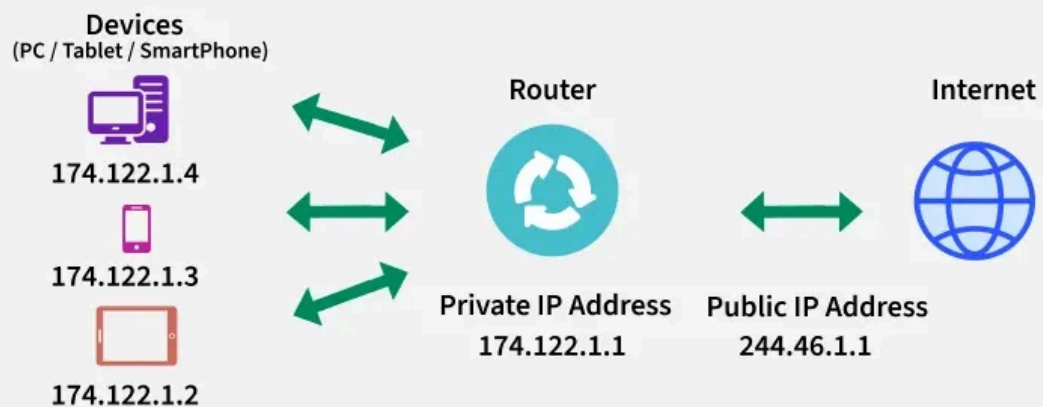


6. Logical Subnetting

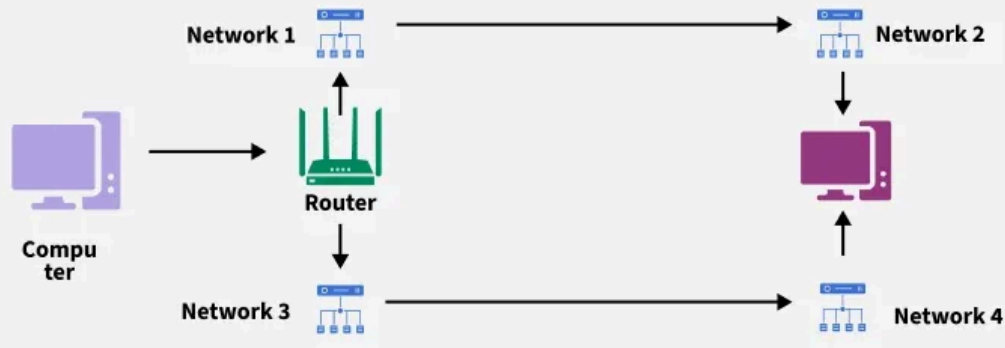


7. Network Address Translation

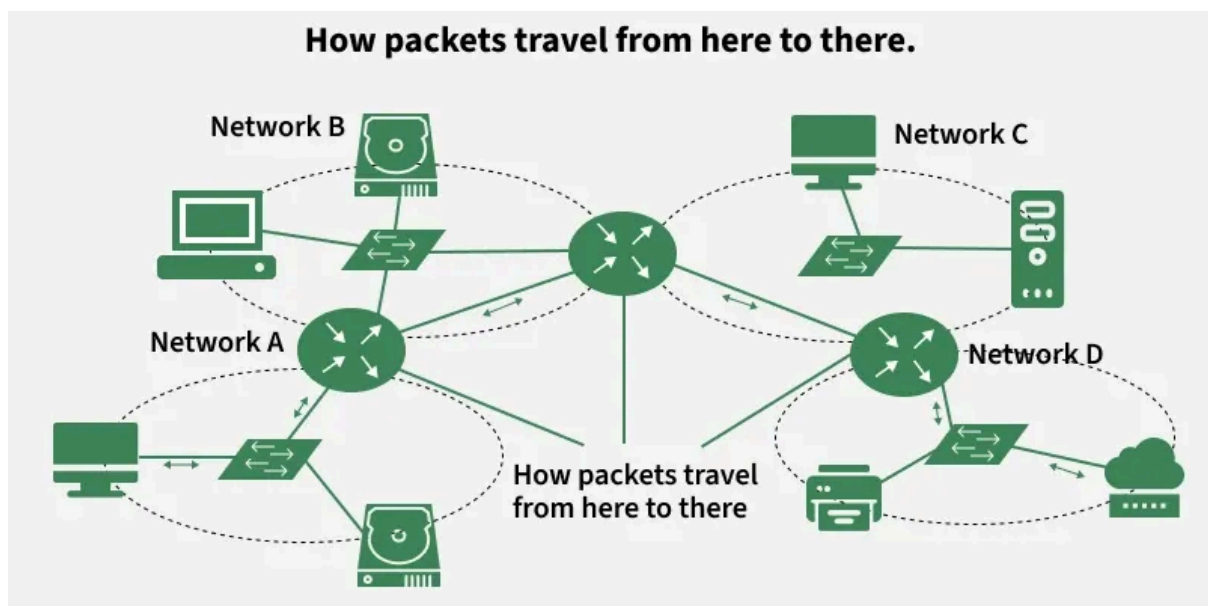
NAT is a method of mapping private IP addresses within a local network to a single public IP address (and vice versa)



8. Routing



How the Network Layer Works



- Each device is assigned a unique logical address (IP address).
- Data from the transport layer is encapsulated into packets, with source and destination IPs attached.
- Routers analyze the destination address and determine the best available path.
- Packets traverse the network hop-by-hop, moving across routers until reaching the destination.
- If the packet size exceeds the MTU, it is fragmented into smaller units.
- At the destination, the fragments are reassembled into the original data.
- If errors occur (e.g., unreachable destination), protocols like ICMP send error messages back to the source.

Protocols Operating at the Network Layer

- IP (Internet Protocol – IPv4/IPv6): Provides logical addressing and delivers packets across networks; IPv6 offers a larger address space and better efficiency.
- ICMP (Internet Control Message Protocol): Sends error reports and diagnostic messages (e.g., destination unreachable, ping).
- ARP (Address Resolution Protocol): Maps IP addresses to MAC addresses within a local network.
- RARP (Reverse Address Resolution Protocol): Retrieves a device's IP address using its MAC address (largely obsolete).
- NAT (Network Address Translation): Converts private IP addresses to public IPs, conserving addresses and improving security.
- IPSec (Internet Protocol Security): Secures IP communication through encryption and authentication.
- MPLS (Multiprotocol Label Switching): Uses labels to forward packets efficiently and manage traffic.

Routing Protocols

- RIP (Routing Information Protocol): Distance-vector protocol using hop count to select routes.
- OSPF (Open Shortest Path First): Link-state protocol that computes the shortest path using network topology.
- BGP (Border Gateway Protocol): Path-vector protocol that routes data between autonomous systems on the internet.

Advantages of the Network Layer

- Enables end-to-end communication across multiple networks.
- Supports scalability by allowing subnetting and hierarchical addressing.
- Efficiently routes packets using shortest-path and dynamic routing algorithms.
- Provides inter-networking by connecting heterogeneous networks.

Limitations of the Network Layer

- No flow control mechanism; congestion may occur if too many datagrams are in transit.
- Limited error control; mainly relies on upper layers for reliability.
- Routers may drop packets under heavy load, leading to possible data loss.
- Fragmentation increases processing overhead and may affect performance.

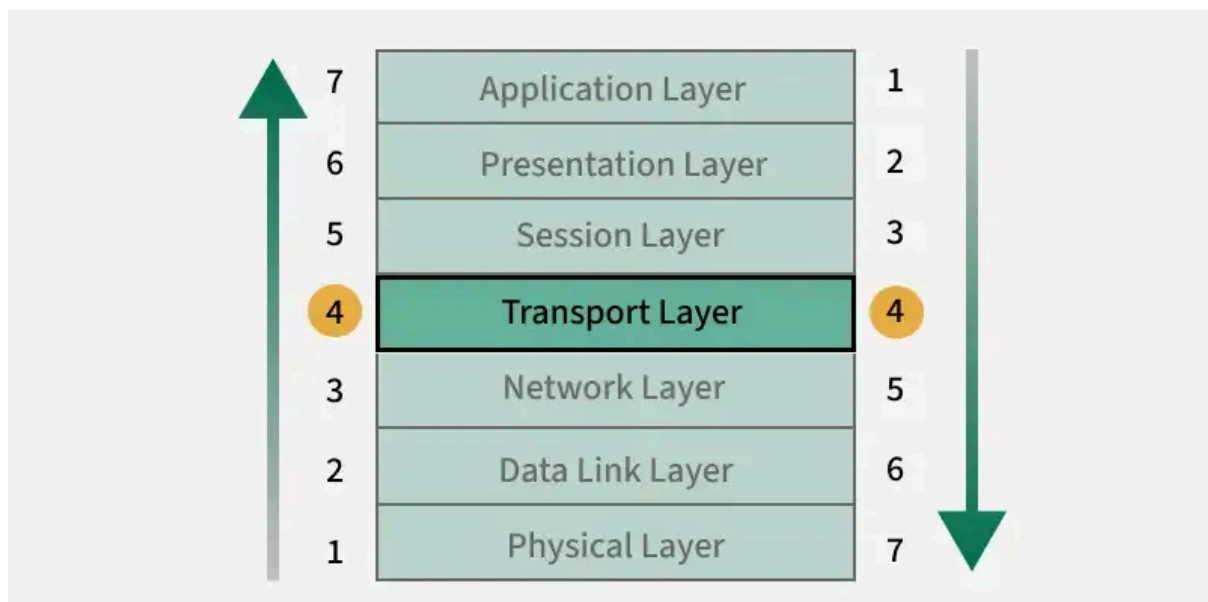
Difference Between Routing and Flooding

Routing	Flooding
A routing table is required.	No routing table is required.
May give the shortest path.	Always gives the shortest path.
Routing is less reliable.	Flooding is more reliable.
Traffic is less in routing.	Traffic is more in flooding.
Duplicate packets are not present.	Duplicate packets are present.

6.Transport Layer in OSI Model

The Transport Layer is the OSI Layer 4 responsible for providing reliable, efficient, and ordered end-to-end communication between applications on different hosts.

- Operates at Layer 4 (Transport Layer) of the OSI model
- Enables process-to-process communication using port numbers
- Ensures reliability through error control, sequencing, and retransmission
- Supports flow control to prevent receiver overload
- Uses protocols like TCP, UDP, and SCTP for different application needs



Functions of Transport Layer

The Transport Layer is responsible for end-to-end communication of data packets. It provides a number of important functions that are responsible for reliable, efficient, and organized data transfer between host systems in a networked environment.

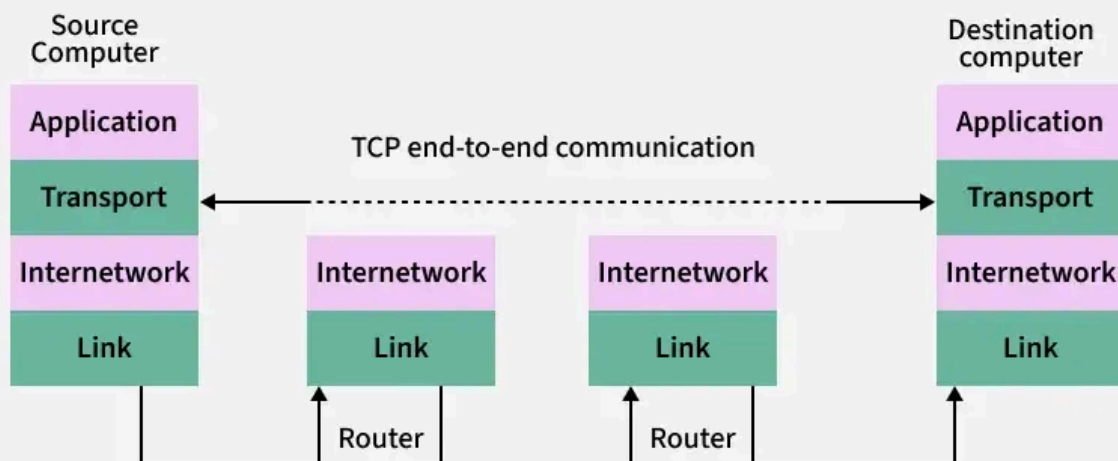
The primary functions of the Transport Layer are:

- Provides end-to-end data delivery
- Ensures reliable and ordered data transfer
- Performs error detection and recovery
- Controls data flow to prevent receiver overload
- Supports multiplexing and demultiplexing using port numbers

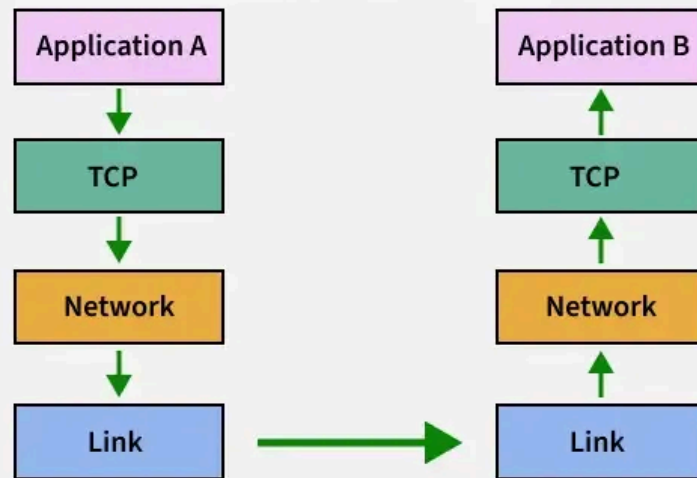
Functions of Transport Layer

1. End-to End Communication
2. Flow Control
3. Multiplexing and Demultiplexing
4. Connection Establishment
5. Connection Termination
6. Reliable Data Delivery
7. Quality of Service(QoS)

1. End-to-End communication

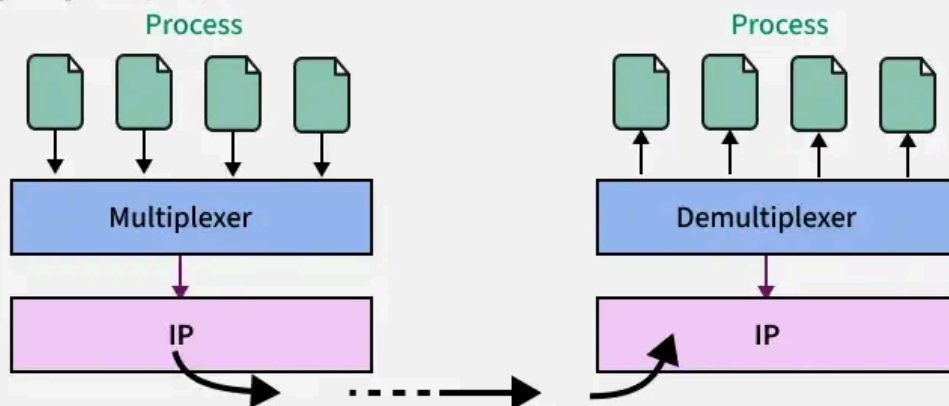


2. Flow Control

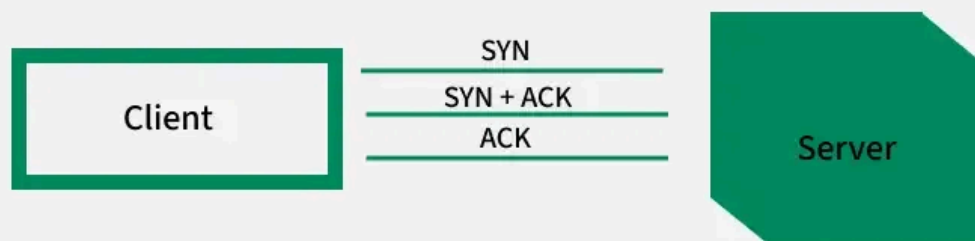


3. Multiplexing and Demultiplexing

It manages multiple data streams on the same network connection by assigning unique ports.

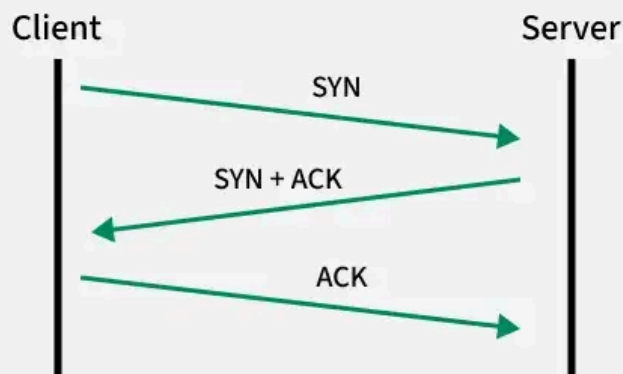


4. Connection Establishment



TCP 3 Way Handshake

5. Connection Termination



6. Reliable Data Delivery

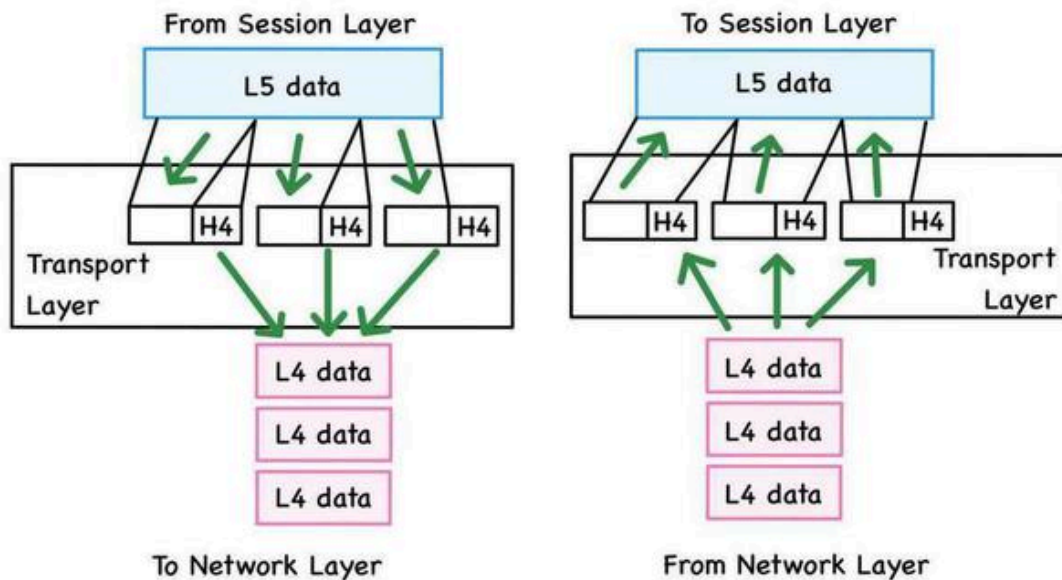
It ensures all data is delivered correctly, in sequence, and without duplication (in protocols like TCP).

7. Quality of Service (QoS) (Optional)

It ensures priority and performance for specific types of data, like video calls or file transfers.

Working of Transport Layer

The Transport Layer provides logical communication between processes on different hosts meaning that even though data travels across various physical networks, the communicating applications perceive a direct, reliable link.



- Implemented only in end systems, not in intermediate routers.
- Uses port numbers to identify sending and receiving applications.
- Supports process-to-process delivery, enabling multiple applications to share a single network connection.
- Performs multiplexing and demultiplexing using port numbers to direct data to the correct process.
- Divides data from upper layers into segments (TCP) or datagrams (UDP) and adds necessary headers.
- Handles error detection, retransmission, and sequencing to maintain reliable communication.
- Coordinates flow control to ensure the receiver is not overloaded.
- Communicates with the Network Layer, which handles addressing and routing, to send data across networks.
- At the receiving end, it removes headers, reassembles data, and passes it to the appropriate application.

3-Way Handshake

The 3-Way Handshake ensures both client and server are ready before data transmission begins:

Step 1: SYN (Client -> Server)

- Client sends a TCP segment with SYN=1, including its ISN (Initial Sequence Number).
- Marks the request to initiate a connection

Step 2: SYN-ACK (Server -> Client)

- Server replies with SYN=1 & ACK=1, containing its own ISN and ACK = client_ISN + 1.
- Confirms receipt of client's SYN and initiates its own sync

Step 3: ACK (Client -> Server)

- Client sends an ACK=1 segment with ACK = server_ISN + 1.
- Completes synchronization; connection enters ESTABLISHED state

Why not just two steps?

With only SYN and SYN-ACK, neither the client nor the server can be sure that the other has successfully received the messages. The third ACK provides mutual confirmation, ensuring both sides are ready before data transfer begins.

Transport Layer Protocols

Transport Layer Protocol uses different protocol for the better communication between two ends uses of protocol may differ from specifications. Below mention are some protocols used in Transport Layer

1. Transmission Control Protocol(TCP)

- TCP is connection-oriented Protocol.
- TCP is reliable protocol.
- As TCP is connection-oriented protocol, so first the connection is established between two ends and then data is transferred and then the connection is terminated after all data being sent.

2. User Datagram Protocol (UDP)

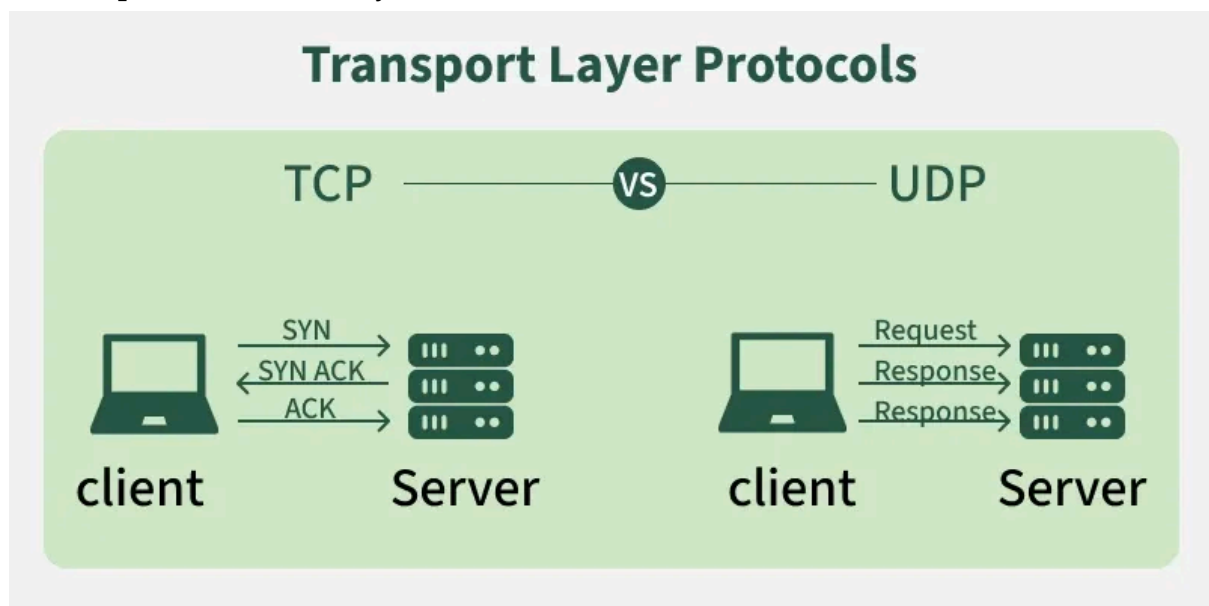
- UDP is not reliable protocol
- The protocol UDP is connectionless.
- When speed and size are more important than security and dependability, this kind of protocol is employed.
- The data from the higher layer is supplemented with transport-level addresses, checksum error control, and length information by UDP, an end-to-end transport level protocol.
- A user datagram is the packet that the UDP protocol generates.

3. Stream Control Transmission Protocol (SCTP)

- Many Internet applications use SCTP to perform transport layer duties, similar to User Datagram Protocol (UDP) and Transmission Control Protocol (TCP).
- On top of a connectionless packet network like IP, SCTP is a dependable transport protocol that facilitates data transfer over the network in scenarios involving one or more IP addresses.

Difference Between TCP and UDP at Transport Layer

Choosing between TCP and UDP depends on the application's need for reliability versus speed and efficiency.

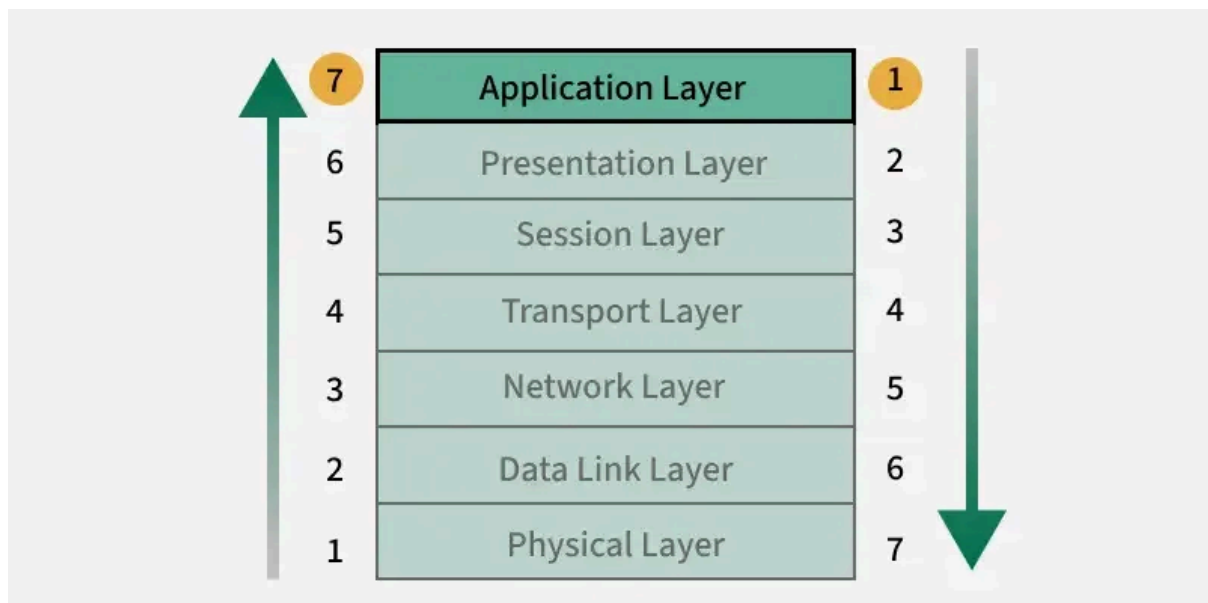


TCP	UDP
TCP is a connection-oriented protocol	UDP is a connection-less protocol
TCP supports error-checking mechanisms	UDP has only the basic error-checking mechanism using checksums
An acknowledgment segment is present	No acknowledgment segment
TCP is slower than UDP	UDP is faster, simpler, and more efficient than TCP
Retransmission of lost packets is possible in TCP, but not in UDP	There is no retransmission of lost packets in UDP
TCP has a (20–60) bytes variable length header	The header length is fixed of 8 bytes

7.Application Layer in OSI Model

The Application Layer is the topmost layer (Layer 7) of the OSI Model that enables direct interaction between end-user applications and the network.

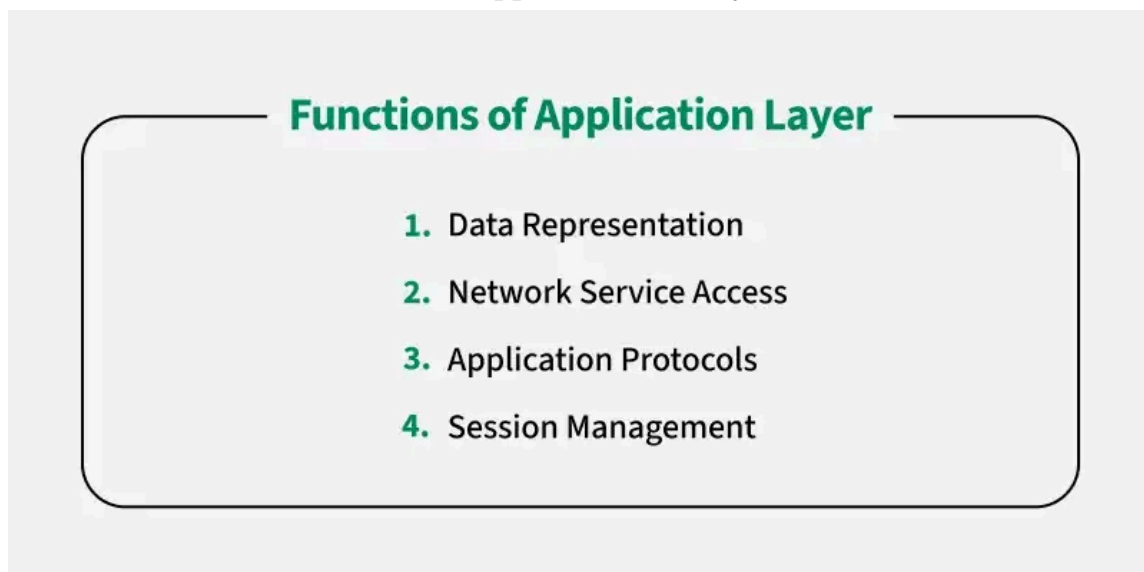
- Closest layer to the end user
- Provides network services to applications
- Enables file transfer, email, web access, and remote login
- Acts as an interface between user software and the network



Note: It provides essential services such as file transfer, remote login, electronic mail, directory services, name resolution and network resource sharing, making it indispensable for modern communication.

Functions of Application Layer

As the highest layer, the Application Layer performs several functions to ensure smooth communication between applications and systems:



1. Data Representation

It ensures that data is in a readable format for both the sender and receiver, including data translation, compression, and encryption.



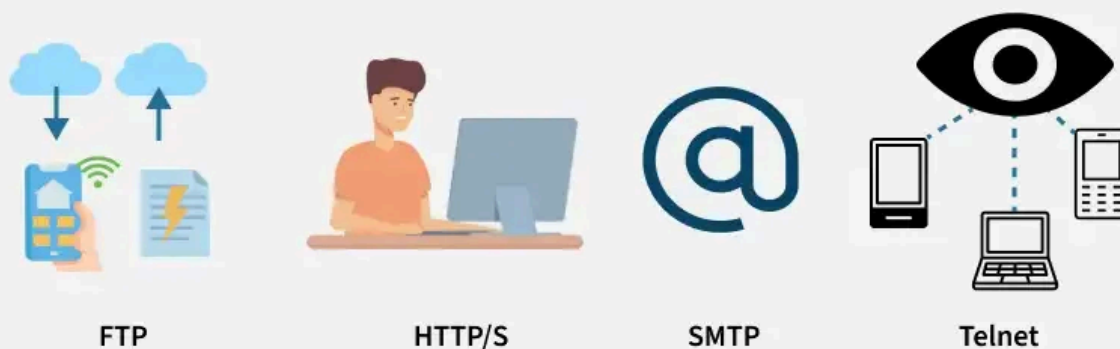
2. Network Service Access

It provides applications with access to network services, enabling communication over a network.



3. Application Protocols

It supports protocols like HTTP, FTP, SMTP, and DNS that enable communication between applications.



4. Session Management

It manage establishment, maintenance, and termination of communication sessions between applications.



1. Data Representation

- Ensures that data is presented in a format that both the sender and receiver can understand.
- Converts user-entered information into a standard format before transmission and interprets received data back into a user-friendly form.
- Example: Converting text, images or files into network-compatible formats like ASCII, JPEG or HTML.

2. Network Service Access

- Provides access to various network services directly to the user.
- Allows applications like email, file transfer and remote login to interact with the underlying network.
- Example: Using email clients (Outlook, Gmail) to access mail services or using FTP to download/upload files.

3. Application Protocols

- Defines a set of rules and procedures that applications use to communicate over a network.
- These protocols handle message formatting, data exchange and service requests.
- Example: HTTP for web browsing, FTP for file transfer, SMTP for emails, DNS for domain name resolution.

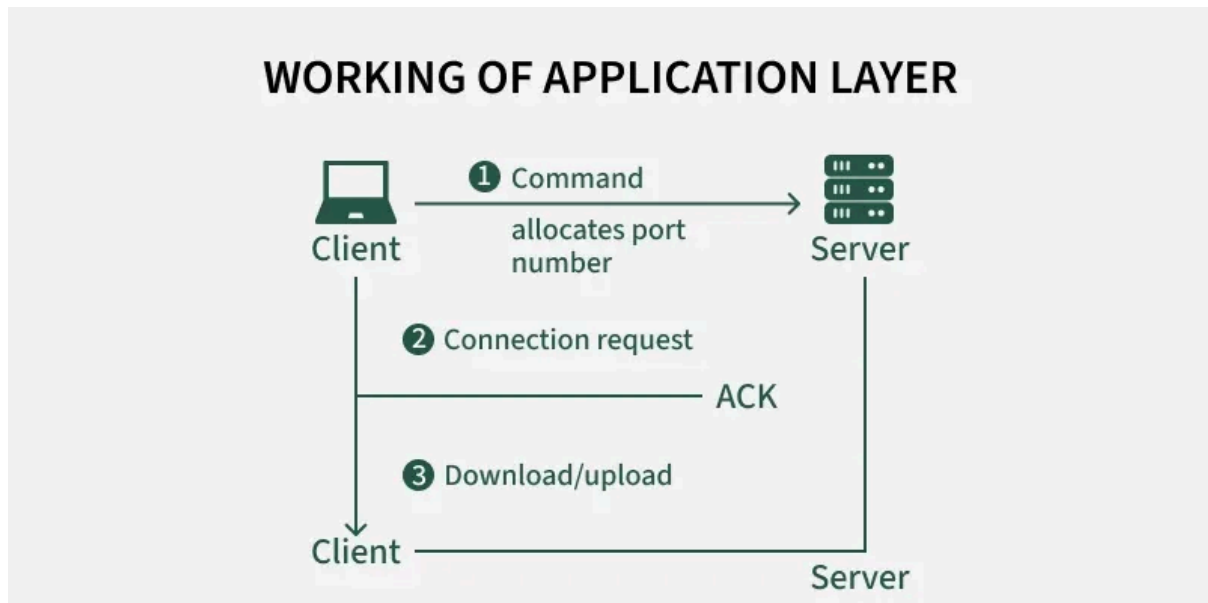
4. Session Management

- Establishes, manages and terminates communication sessions between applications.

- Maintains synchronization during data exchange and ensures a proper connection is closed after the communication ends.
- Example: Logging into a remote server using Telnet or SSH, where a session is created and terminated after logout.

Working of Application Layer

Once the data exchange is complete, the client or server can terminate the session and the connection is closed gracefully.



- At first, client sends a command to server and when server receives that command, it allocates port number to client.
- Thereafter, the client sends an initiation connection request to server and when server receives request, it gives acknowledgement (ACK) to client through which client has successfully established a connection with the server.
- Therefore, now client has access to server through which it may either ask server to send any types of files or other documents or it may upload some files or documents on server itself.

Services Provided by Application Layer Protocols

The following are some of the services which are provided by Application layer protocols-

- The Application Layer protocol defines process for both parties which are involved in communication.
- These protocols define the type of message being sent or received from any side (either source host or destination host).
- These protocols also define basic syntax of the message being forwarded or retrieved.

- These protocols define the way to send a message and the expected response.
- These protocols also define interaction with the next level.

Protocols of the Application Layer

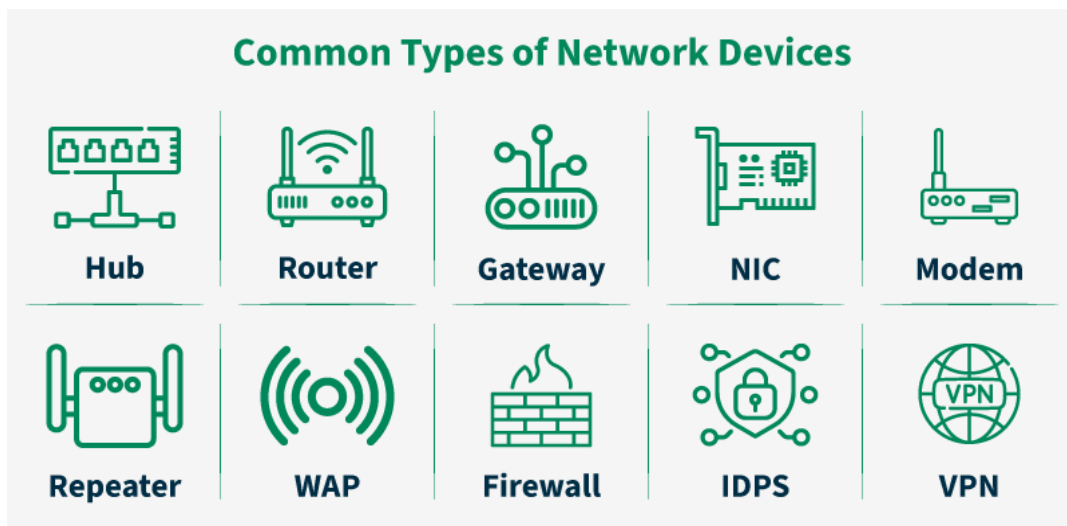
The application layer provides several protocols which allow any software to easily send and receive information and present meaningful data to its users. The following are some of the application layer protocols.

- **HTTP (HyperText Transfer Protocol):** Used for web communication (Port 80).
- **DNS (Domain Name System):** Translates domain names into IP addresses (Port 53).
- **TELNET:** Provides remote login and file management services (Port 23).
- **DHCP (Dynamic Host Configuration Protocol):** Assigns IP addresses to hosts dynamically (Ports 67 & 68).
- **FTP (File Transfer Protocol):** Facilitates file transfer between systems (Ports 20 for data, 21 for control).
- **SMTP (Simple Mail Transfer Protocol):** Handles sending of emails (Ports 25 & 587).
- **NFS (Network File System):** Enables remote access to files as if they were local (Port 2049).
- **SNMP (Simple Network Management Protocol):** Manages and monitors devices in a network (Ports 161 & 162).

8. Network Devices

Network Devices are the physical appliances required for communication and interaction between computers on a computer network.

- Enable communication by transmitting and receiving data between devices.
- Allow devices to connect to networks efficiently and securely.
- Improve network performance by reducing congestion and managing traffic.
- Extend network coverage and solve signal loss or attenuation problems.



Layer 1 Devices: Physical Layer

These devices deal with raw electrical or optical signals (bits). These devices do not understand IP addresses or MAC addresses.

1. Hub

It is a central connection point for devices in a LAN.

- It takes an incoming signal on one port and blindly broadcasts it to all other ports without knowing the recipient.
- High network traffic collisions and security risks (everyone sees everyone's data).
- It is an obsolete device, replaced by Switches.

2. Repeater

It regenerates signals to extend the range of a network.

- It receives a weak signal (due to attenuation over long cables), amplifies it, and retransmits it.
- Extending WiFi range or Ethernet cables beyond 100 meters.

3. Modem (Modulator-Demodulator)

This bridges the gap between digital computer data and analog signals, allowing devices to connect to the internet via telephone, cable, or fiber lines.

- Connects your home network to the ISP (Internet Service Provider).
- Converts digital signals from your computer into analog signals for telephone/cable lines (Modulation) and vice-versa (Demodulation).

Layer 2 Devices: Data Link Layer

These devices work with MAC Addresses (Physical Addresses). They are smarter than hubs.

4. Switch

A switch is a high-speed networking device that connects devices (computers, printers, servers) within a Local Area Network (LAN),

- Unlike hub, switch learns the MAC address of every connected device.
- It sends data only to the specific port where the destination device is connected.
- Reduces collisions and improves security and speed.

5. Bridge

This connects two separate LAN segments to make them appear as one.

- Filters traffic based on MAC addresses to keep local traffic local, reducing congestion.
- Largely replaced by Switches (which are essentially multi-port bridges).

6. Access Point (AP)

It is a networking hardware device that allows Wi-Fi-enabled devices, such as laptops and smartphones, to connect to a wired network.

- Acts as a bridge between wired Ethernet and wireless Wi-Fi devices.
- An AP only provides Wi-Fi signal and it does not route traffic or assign IP addresses.

Layer 3 Devices: Network Layer

These devices work with IP Addresses (Logical Addresses) and connect different networks together.

7. Router

It is a networking device that connects multiple networks and directs data packets between them using IP addresses.

- It uses IP Addresses to determine the best path for data packets to travel.
- It maintains a Routing Table to make these decisions.
- Routers stop broadcast traffic, isolating networks from each other.

8. Brouter (Bridging Router)

It is a networking device combining bridge and router functions, operating at both the Data Link and Network layers of the OSI model.

- It can route known protocols (like TCP/IP).
- It bridge unknown protocols (like non-routable legacy traffic) at the same time and rarely used today.

Layer 4-7 Devices: Advanced Processing

9. Gateway

It is a hardware device or software node that acts as an entry/exit point between two distinct networks using different protocols, translating data to ensure compatibility.

- It can convert protocols, data formats, or architectures (e.g., connecting a TCP/IP network to a legacy Mainframe network).
- An Internet Gateway translates your private LAN requests into public internet requests.

10. Firewall

It is a network security system, comprising hardware, software, or both that monitors and controls incoming and outgoing network traffic based on predetermined security rules

- It is a physical appliance at the network edge.
- This is installed on a specific server or PC.
- It tracks active connections.

Hub vs. Switch vs. Router

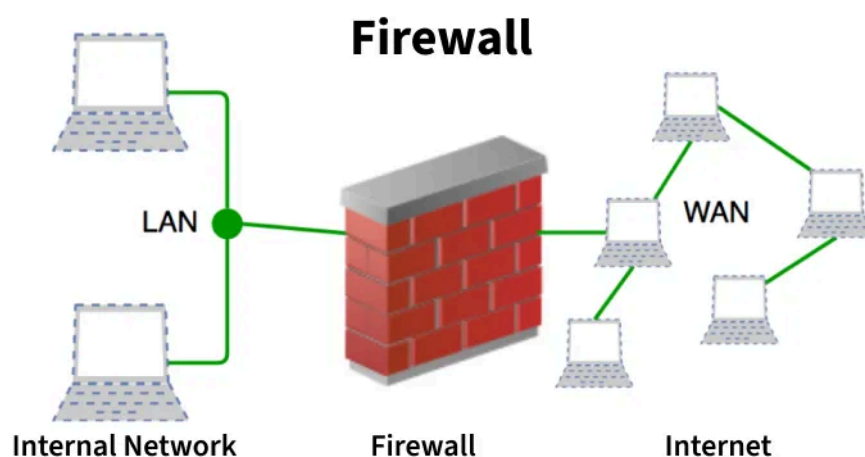
Feature	Hub	Switch	Router
Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Type	None (Bits)	MAC Address	IP Address
Data Flow	Broadcast (to all)	Unicast (to destination)	Route (best path)
Intelligence	Dumb	Smart	Smarter
Used For	Obsolete	Connecting Devices (LAN)	Connecting Networks (WAN)

9.Security Basics

9.1 Introduction of Firewall in Computer Network

A firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to either:

- Accept: Allow the traffic.
- Reject: Block with an error response.
- Drop: Block silently without response.



Importance of Firewalls

A firewall is the first line of defense in cybersecurity, acting as a security barrier between internal systems and external networks. It forces all traffic through a single checkpoint, where data packets are monitored, filtered, and either allowed or blocked based on predefined rules. Firewalls are essential because they:

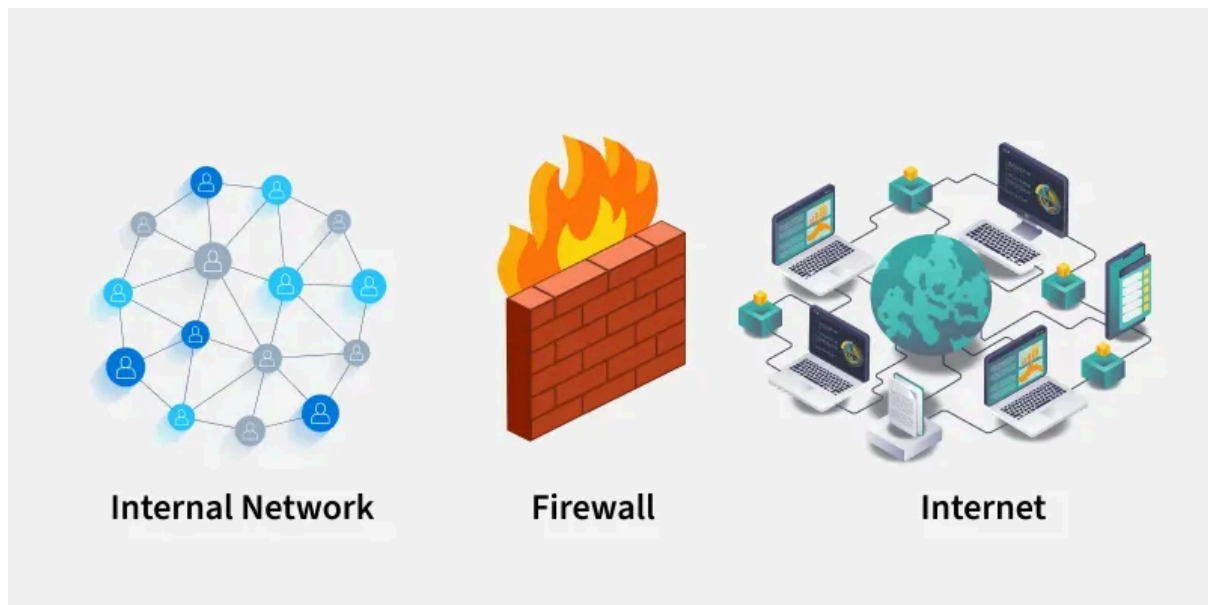
- **Prevent Unauthorized Access:** Like a locked door with a guard, only trusted users and traffic are allowed through.
- **Block Malicious Traffic:** Harmful data such as viruses, phishing attempts, or denial-of-service (DoS) attacks are stopped before reaching the system.
- **Protect Sensitive Information:** Safeguards personal and business data from theft or accidental leaks.
- **Control Network Usage:** Enforces policies such as parental controls, workplace restrictions, or government filtering.
- **Mitigate Insider Risks:** Detects suspicious applications or data exfiltration attempts from within the network.

By combining prevention, monitoring, and control, firewalls provide customizable and comprehensive protection against both external and internal threats.

Working of Firewall

A firewall inspects all incoming and outgoing traffic and decide whether to allow or block it.

1. All data packets entering or leaving the network must first pass through the firewall.
2. The firewall examines each packet against predefined security rules set by the organization.
3. If the packet matches safe rules, it is allowed; if it is suspicious, blacklisted, or contains malicious content, it is blocked.
4. Blocked or unusual traffic is recorded in logs, and real-time alerts may be generated for serious threats.
5. Since it is not possible to define every rule, the firewall applies a default policy (accept, reject, or drop). Setting the default policy to drop or reject is considered best practice to prevent unauthorized access.



Default Policy: A firewall needs a default action (accept, reject, or drop) for traffic not covered by rules. For example, if no rule exists for SSH, the default applies. To prevent unauthorized access, it is best set to drop or reject.

Types of Firewall

Firewalls can be categorized based on their generation.

1) Network Placement

- Packet Filtering Firewall
- Stateful Inspection Firewall
- Proxy Firewall (Application Level)
- Circuit-Level Gateway
- Web Application Firewall (WAF)
- Next-Generation Firewall (NGFW)

2) Systems Protected

- Network Firewall
- Host-Based Firewall

3) Data Filtering Method

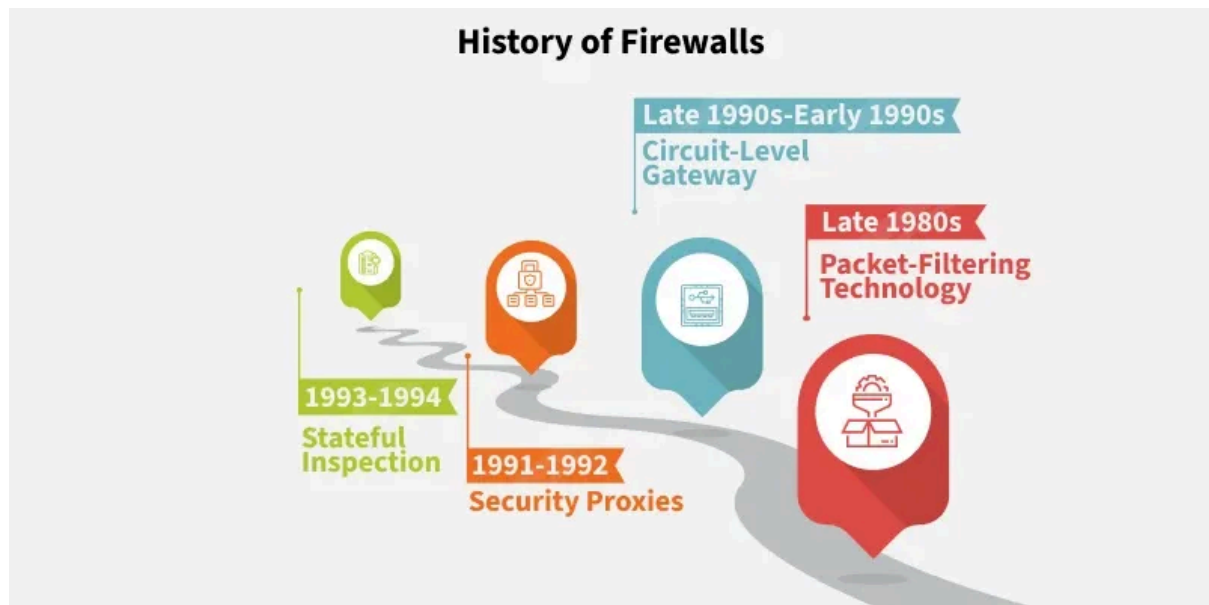
- Perimeter Firewall
- Internal Firewall
- Distributed Firewall

4) Form Factors

- Hardware Firewall
- Software Firewall

History of Firewalls

Firewalls evolved from simple packet filtering to advanced, user-friendly security systems used worldwide.



- Late 1980s: Jeff Mogul, Brian Reid, and Paul Vixie at Digital Equipment Corp (DEC) developed packet-filtering technology, laying the groundwork for firewalls by checking external connections before they reached internal networks.
- Late 1980s(Early 1990s): AT&T Bell Labs researchers, including Presotto, Sharma, and Nigam, developed the circuit-level gateway, a firewall that vetted ongoing connections without reauthorizing each data packet, paving the way for more efficient security.
- 1991-1992: Marcus Ranum introduced security proxies at DEC, leading to the creation of the Secure External Access Link (SEAL), the first commercially available application-layer firewall, based on earlier DEC work.
- 1993-1994: At Check Point, Gil Shwed pioneered stateful inspection technology, filing a patent in 1993. Nir Zuk developed a graphical interface for Firewall-1, making firewalls accessible and widely adopted by businesses and homes.

9.2 What is Data Encryption?

Data encryption is the process of converting readable information (plaintext) into an unreadable format (ciphertext) to protect it from unauthorized access. It is a method of preserving data confidentiality by transforming it into ciphertext, which can only be decoded using a unique decryption key produced at the time of the encryption or before it. The conversion of plaintext into ciphertext is known as encryption. By using encryption keys and mathematical algorithms, the data is scrambled so that anyone intercepting it without the proper key cannot understand the contents.

When the intended recipient receives the encrypted data, they use the matching decryption key to return it to its original, readable form. This approach ensures that sensitive information such as personal details, financial data, or confidential communications remains secure as it travels over networks or is stored on devices.

Key Objective of Encryption Data

- **Confidentiality:** Encryption ensures that only authorized parties can get access to data and recognize the information.
- **Data Integrity:** Encryption can also provide data integrity by making sure that the encrypted data remains unchanged during transmission. Any unauthorized changes to the encrypted information will render it undecipherable or will fail integrity checks.
- **Authentication:** Encryption may be used as part of authentication mechanisms to verify the identification of the communication party.
- **Non-Repudiation:** Through encryption, events can make sure that they cannot deny their involvement in growing or sending a selected piece of data.

Importance of Data Encryption

The significance of encryption cannot be overstated in any way. Even though your data is stored in a standard infrastructure, it is still possible for it to be hacked. There's always the chance that data will be compromised, but with data encryption, your information will be much more secure. Consider it this way for a moment. If your data is stored in a secure system, encrypting it before sending it out will keep it safe. Sanctioned systems do not provide the same level of protection.

So, how do you think this would play out in real life?

Suppose the user has access to sensitive information while at work. The user may put the information on a portable disc and move it anywhere they choose without any encryption. If the encryptions are set in place ahead of time, the user can still copy the information, but the data will be unintelligible when they try to see it somewhere else. These are the benefits of data encryption that demonstrate its genuine value.

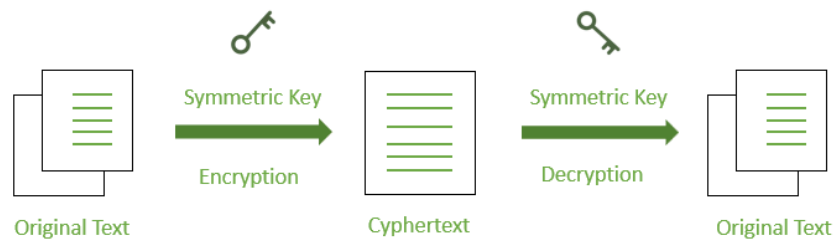
Types of Data Encryption

There are multiple encryption techniques, each of which have been developed with various security requirements in mind. Symmetric and Asymmetric encryption are the two types of data encryption.

1. Symmetric Key Encryption

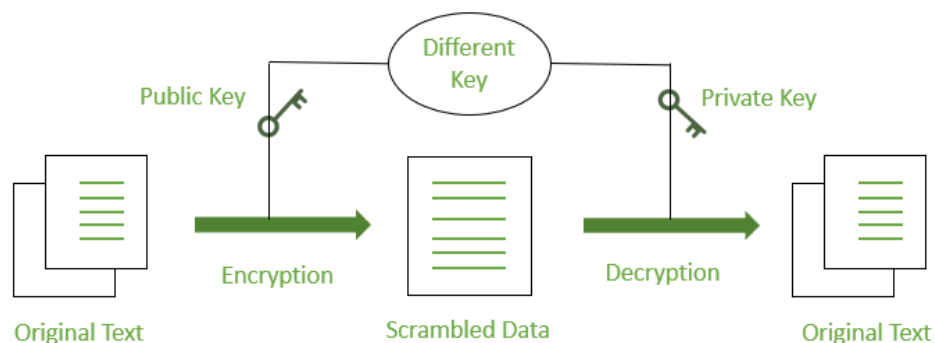
There are a few strategies used in cryptography algorithms. For encryption and decryption processes, some algorithms employ a unique key. In such operations, the unique key must be secured since the system or person who knows the key has

complete authentication to decode the message for reading. This approach is known as "symmetric encryption" in the field of network encryption.



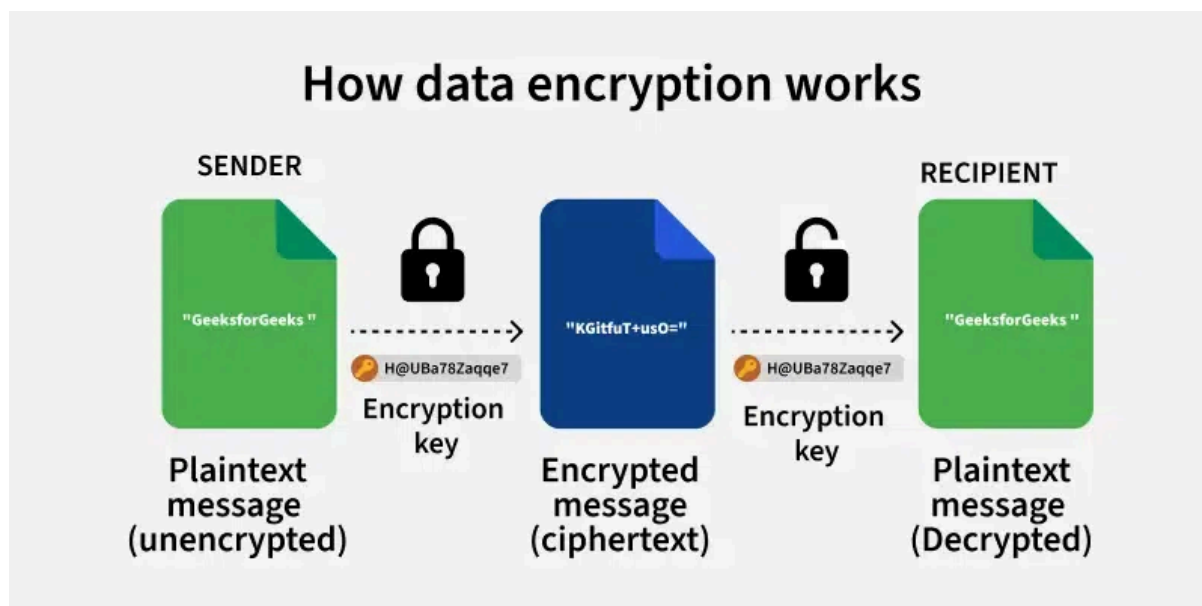
2. Asymmetric Key Encryption

Some cryptography methods employ one key for data encryption and another key for data decryption. As a result, anyone who has access to such a public communication will be unable to decode or read it. This type of cryptography, known as "public-key" encryption, is used in the majority of internet security protocols. The term "asymmetric encryption" is used to describe this type of encryption.



How Does Encryption Work?

When data or information is shared over internet, it passes via a number of global network devices that are a component of the public internet. Data that is transmitted via the open internet leads to the risk of being stolen or hacked by hackers. Users can install particular hardware or software to guarantee the safe transfer of data or information in order to avoid hacking. In network security these operations are referred to as encryption. The process of transforming plaintext into ciphertext, is called encryption.



On the left you have an original, readable message called plaintext such as "GeeksforGeeks." Before sending it over a network, the sender uses an encryption key and an encryption process to convert this readable message into a scrambled, unreadable format known as ciphertext (in this image it is like "KGifut+usO="). This ciphertext travels across the internet, so if someone intercepts it, they cannot understand it without the key. When the ciphertext reaches the intended recipient, they use the matching decryption key and a decryption process to turn the unreadable ciphertext back into the original, readable message "GeeksforGeeks." Essentially the image shows how encryption and decryption ensure that only authorized parties with the correct keys can access the information in its original form.

States of Data Encryption

Data encryption in transit: Information that is actively moving from one point to another, such as via the internet or over a private network, is referred to as data in transit. Data is deemed less safe when in transit due to the weaknesses of transfer techniques.

Encryption of data at rest: Data encryption at rest decreases the risk of data breach caused by lost or stolen devices, inadvertent password sharing, or accidental permission granting by increasing the time it takes to access information and providing the time required to discover data loss, ransomware attacks, remotely erased data, or changed credentials.

How the Data Encryption Takes Place?

Data encryption transforms readable data known as plaintext, into an unreadable format called ciphertext. This process generally involves an algorithm and a unique

encryption key. The algorithm uses the key to scramble the data in such a way that anyone without the key cannot make sense of the ciphertext.

When the intended recipient receives the encrypted data, they use the corresponding decryption key often related to the encryption key to reverse the process and restore the data to its original readable form. This approach ensures that even if someone intercepts the data during transmission they cannot understand it unless they have the correct key.



Encryption is performed on digital communications, this technological procedure is designed to prevent a third party from deciphering the signal's secret content. Consumers conduct transactions for goods purchases over the internet. There are millions of web services that can help various trained employees do their responsibilities. Furthermore, to utilize these services that demand personal information, most websites require substantial identification. One of the most common ways, known as "encryption," is to keep such information safe and secure.

Uses of Data Encryption

- Using digital signatures, Encryption is used to prove the integrity and authenticity of the information. Digital-rights management and copy protection both require encryption.
- Encryption can be used to erase data. But since data recovery tools can sometimes recover deleted data, if you encrypt the data first and then throw away the key, the only thing anyone can recover is the ciphertext, not the original data.
- Data Migration is used when transferring data over a network to ensure that no one else on the network can read it.
- VPNs (Virtual Private Networks) uses encryption, and you should encrypt everything you store in the cloud. This can encrypt the entire hard drive as well as voice calls.

Advantages of Data Encryption

- Data encryption keeps information distinct from the security of the device on which it is stored. Encryption provides security by allowing administrators to store and send data via insecure channels.

- If the password or key is lost, the user will be unable to open the encrypted file. Using simpler keys in data encryption, on the other hand, makes the data insecure, and anybody may access it at any time.
- Encryption improves the security of our information.

Disadvantages of Data Encryption

- If the password or key is lost, the user will be unable to open the encrypted file. Using simpler keys in data encryption, on the other hand, makes the data insecure, and anybody may access it at any time.
- Data encryption is a valuable data security approach that necessitates a lot of resources, such as data processing, time consumption, and the use of numerous encryption and decryption algorithms. As a result, it is a somewhat costly approach.
- Data protection solutions might be difficult to utilize when the user layers them for contemporary systems and applications. This might have a negative influence on the device's normal operations.
- If a company fails to realize any of the restrictions imposed by encryption techniques, it is possible to set arbitrary expectations and requirements that might undermine data encryption protection.

Data Encryption Algorithms

Depending on the use case, there are a variety of data encryption algorithms to choose from, but the following are the most commonly used:

- DES (Data Encryption Standard) is an old symmetric encryption algorithm that is no longer considered suitable for modern applications. As a result, DES has been superseded by other encryption algorithms.
- Triple DES (3DES or TDES): Encrypts, decrypts, and encrypts again to create a longer key length by running the DES algorithm three times. It may be run with a single key, two keys, or three separate keys to increase security. 3DES is vulnerable to attacks such as block collisions since it uses a block cipher.
- RSA is a one-way asymmetric encryption algorithm that was one of the first public-key algorithms. Because of its long key length, RSA is popular and widely used on the Internet. It is used by browsers to create secure connections over insecure networks and is part of many security protocols such as SSH, OpenPGP, S/MIME, and SSL/TLS.
- Twofish is one of the fastest algorithms, with sizes of 128, 196, and 256 bits and a complex key structure for added security. It is available for free and is included in some of the best free software, including VeraCrypt, PeaZip, and KeePass, as well as the OpenPGP standard.
- Elliptic Curve Cryptography (ECC) was created as an upgrade to RSA and offers better security with significantly shorter key lengths. In the SSL/TLS protocol, ECC is an asymmetric method.

- The Advanced Encryption Standard (AES) is the encryption standard used by the US government. The AES algorithm is a symmetric-key algorithm that employs block cipher methods. It comes in sizes of 128, 192, and 256 bits, with the number of rounds of encryption increasing as the size increases. It was designed to be simple to implement in both hardware and software.

Conclusion

Encryption is a way of turning readable data into a secret code so that only authorized people can access it. It protects important information whether it's being sent from one place to another or stored on a device from being seen by anyone who doesn't have the right key to unlock it.

As we share and store more personal, financial, and business data online, encryption becomes more important. It helps keep our information safe, builds trust, and supports secure communication around the world.